Degeneracy in Astrocytic Potassium Buffering

Chapter 2 — digital supplement: supporting figures of the potassium analyses

This document accompanies Chapter 2 of the thesis. It contains the supporting figures of the potassium analyses (Sections 2.3.9 and 2.3.10 of the thesis) that are not reproduced in the thesis itself. The figures are numbered D1–D69 in the order of their first citation in the chapter. Together with the main figures of Chapter 2 (Figs. 2.12–2.26) and the supporting figures of Appendix C (Figs. C.4–C.16), they reproduce the complete graphical output of the potassium analyses. References to figures numbered 2.x or C.x point to the thesis.

All figures show model output, not independent biological observations. Panel labels are those of the original analyses; the captions translate them into the notation of the chapter (κ, shown as Z_s in some panels; $P_{K,\mathrm{gap}}$, shown as P_{gap}; $g_{K,a}$, the background potassium conductance, shown as g_K; and α_t/α_s, shown as α_1/α_2). A concentration at the end of a finite window, the pre-input concentration and a verified stationary concentration are distinct quantities, and variation among model cells is not animal-level uncertainty. Figures D24–D54 belong to an earlier profile-robustness analysis with a near-complete Kir reduction; they are kept as controls, and the primary mechanism-loss experiment of the chapter is the 50% reduction of effective junctional permeability with Kir conductance retained.

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Cumulative and generic formulations

S2. Recruitment memory and extracellular recovery must be separated

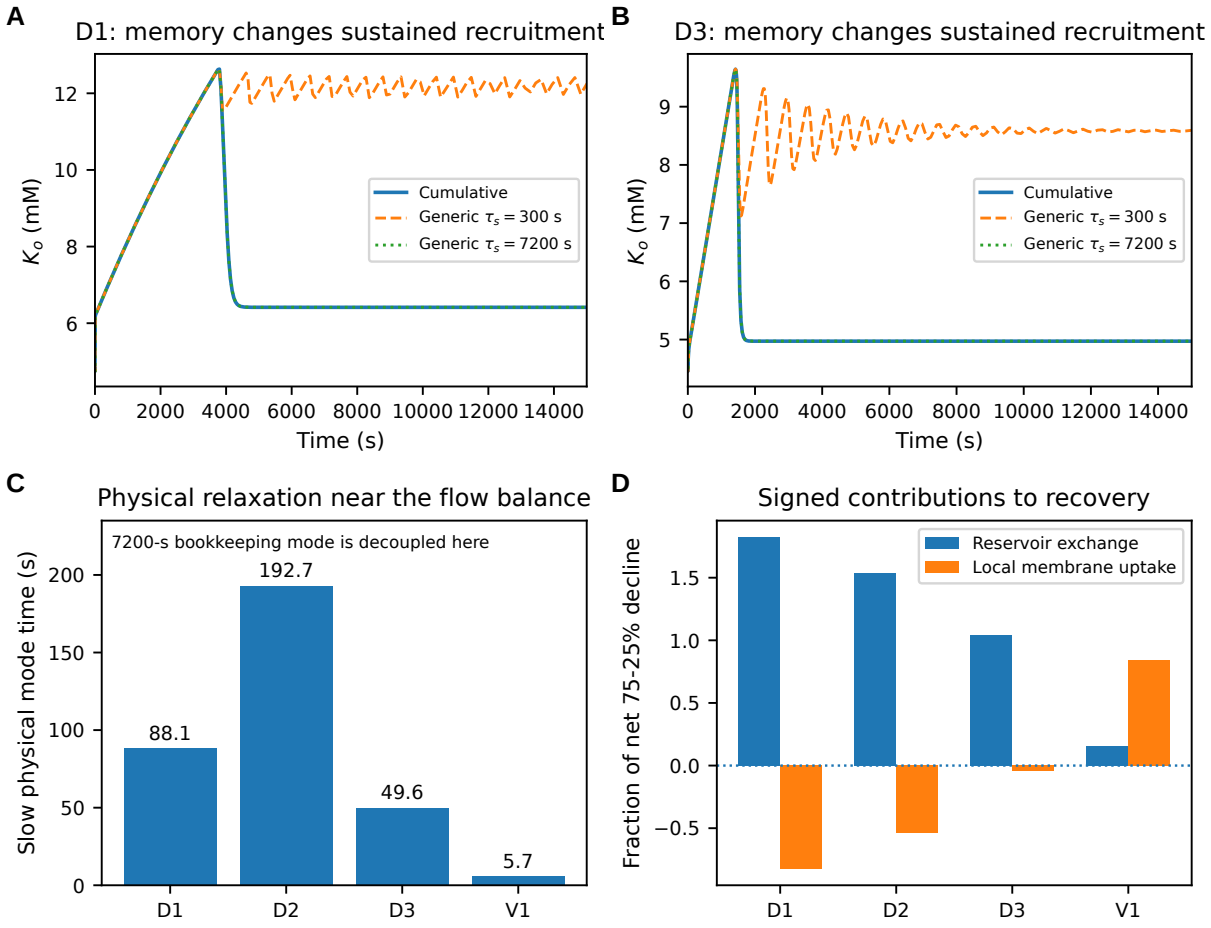


Figure D1: Recruitment memory and extracellular recovery are separate processes. (A, B) D1 and D3 in the cumulative formulation and in the generic formulation with 300-s and 7,200-s memory: the memory changes sustained recruitment. (C) Time constant of the slow physical relaxation near the flow balance in D1, D2, D3 and V1; the 7,200-s memory mode is decoupled from this relaxation. (D) Signed contributions of reservoir exchange and local membrane uptake to the net 75–25% decline of extracellular potassium. A negative contribution indicates that the pathway opposed the decline.

S1. Long-time limiting analysis; not a physiological-duration trace

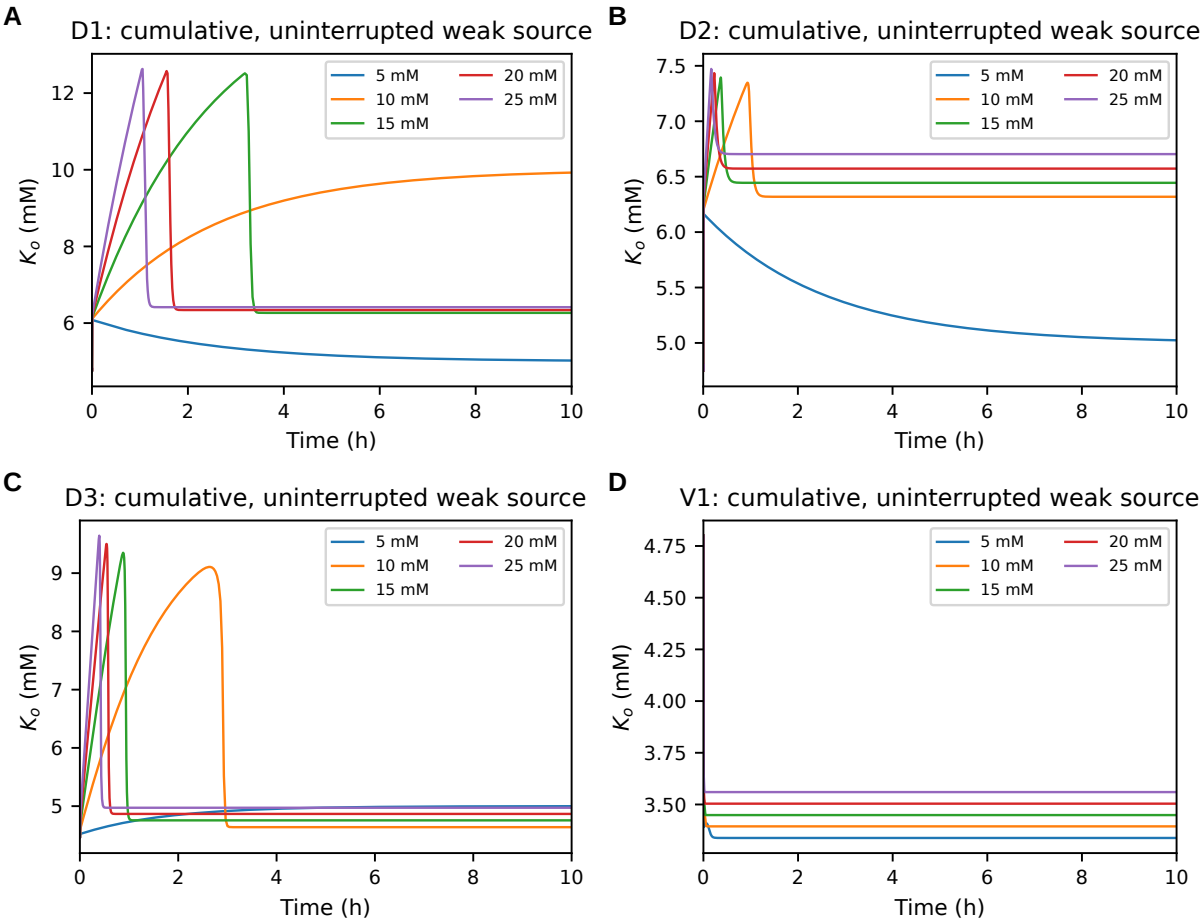


Figure D2: Long-time responses of the cumulative formulation to an uninterrupted weak supply. Extracellular potassium in D1, D2, D3 and V1 (A–D) during 10 h of uninterrupted weak supply from reservoirs of 5–25 mM. This is a limiting analysis of the formulation, not a trace of physiological duration.

S4. Intermediate supply changes timing and end concentration together

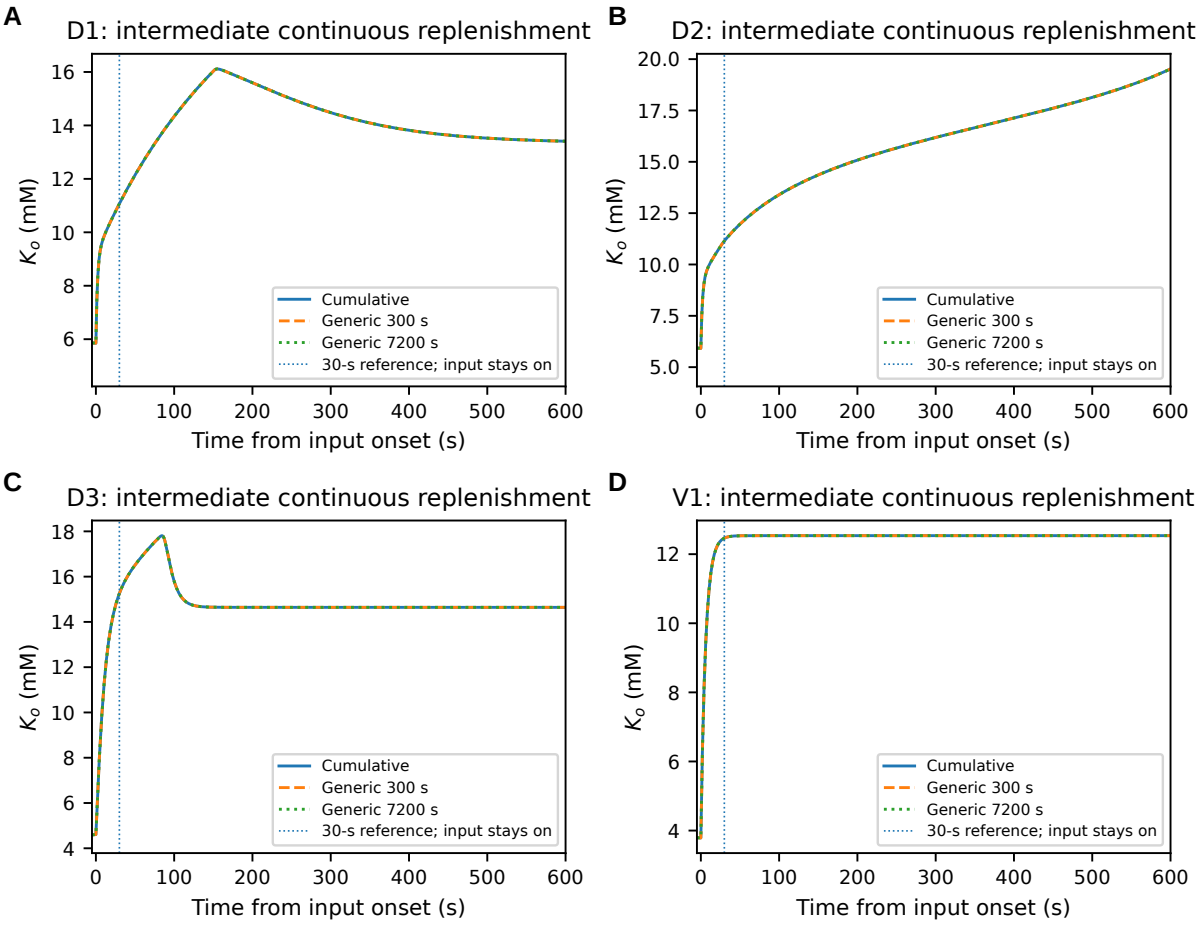


Figure D3: An intermediate continuous supply changes timing and final concentration together. D1, D2, D3 and V1 (A–D) under continuous supply of intermediate strength, in the cumulative formulation and in the generic formulation with 300-s and 7,200-s memory; the 30-s reference is shown with the input maintained. The formulations differ little for this protocol.

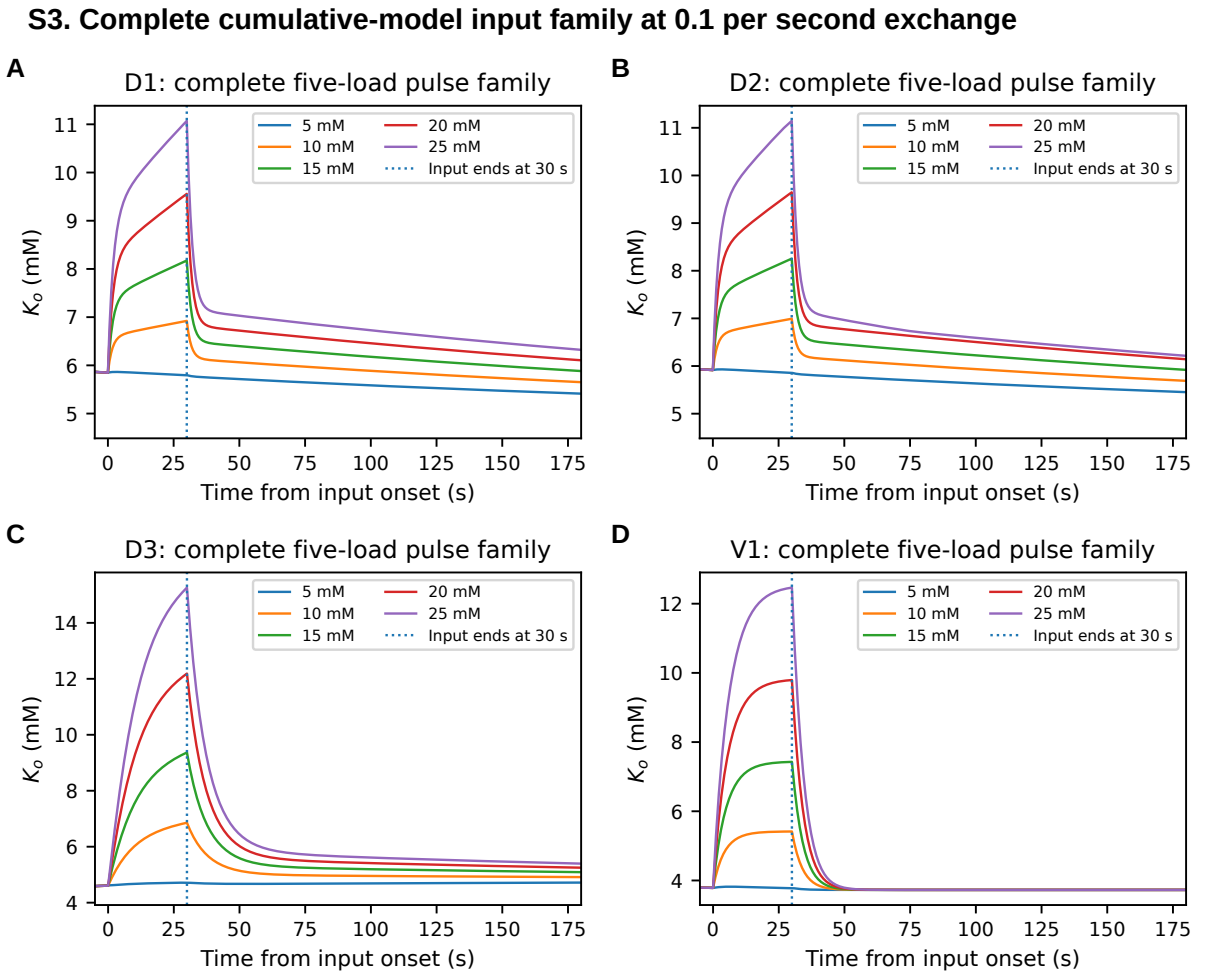


Figure D4: Complete input family of the cumulative formulation at $\epsilon = 0.1\,\mathrm{s}^{-1}$. Extracellular potassium in D1, D2, D3 and V1 (A–D) for 30-s inputs from reservoirs of 5–25 mM, followed by 150 s of observation. The dotted line marks the end of the input.

Potassium responses of the twenty models under CONTROL

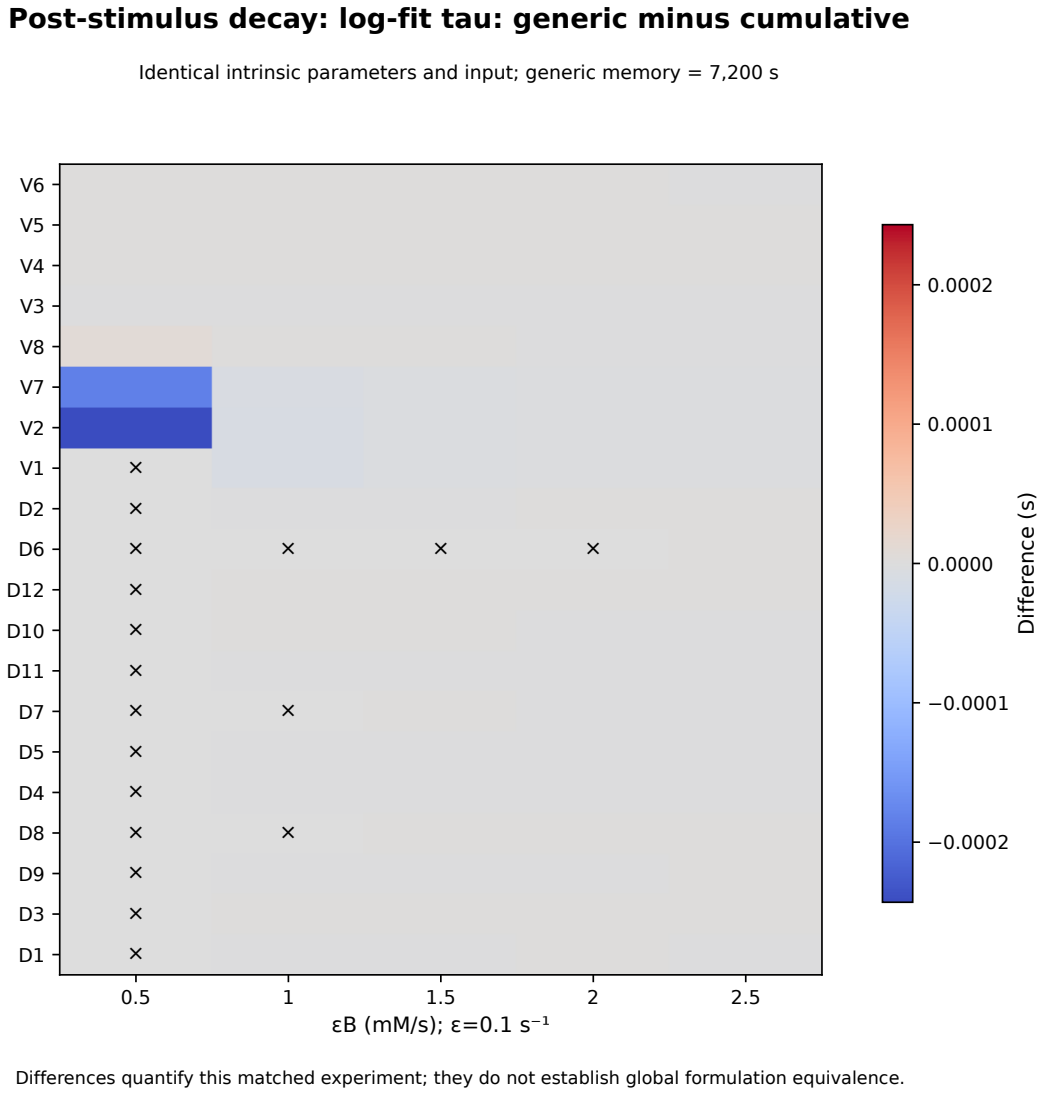


Figure D5: Generic minus cumulative formulation: decay time constant. As in Fig. C.5; the differences are of order 10^{-4} s, and crosses mark loads without a valid fit.

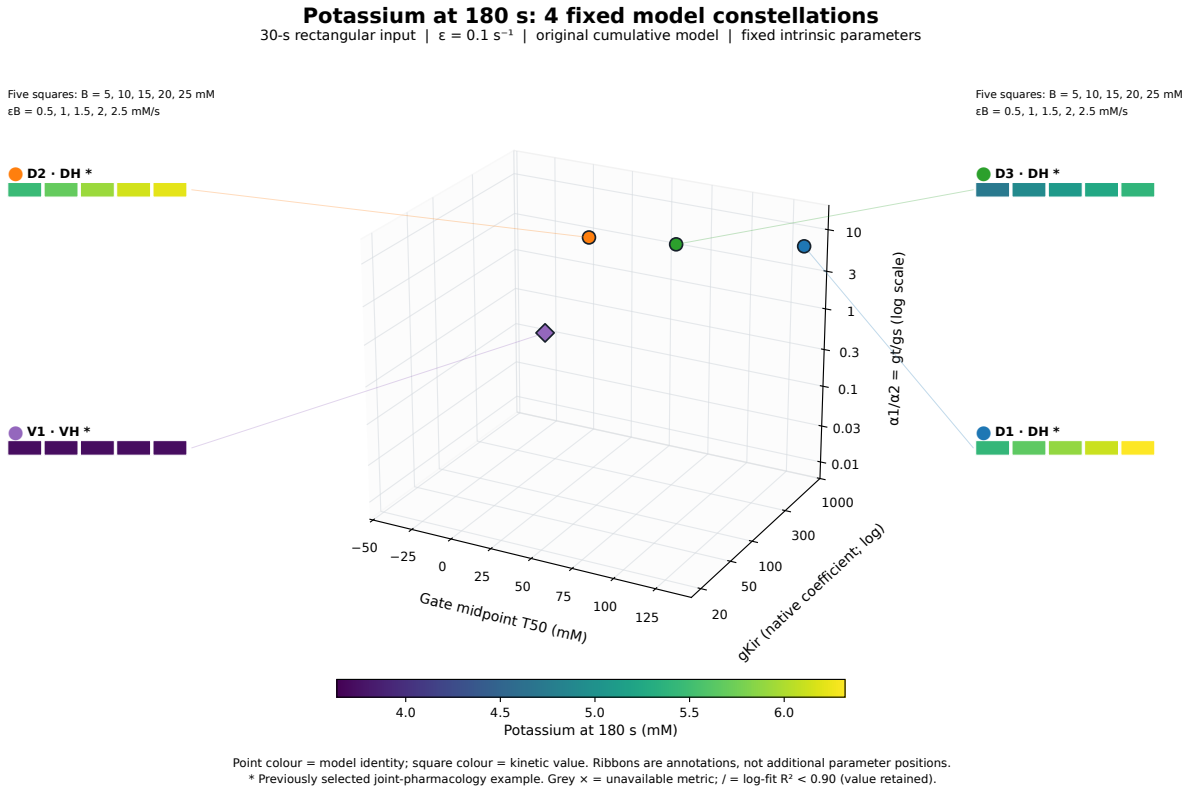


Figure D6: Potassium at 180 s for the four jointly supported models in parameter space. Each model is placed by its gate midpoint T_{50}, its Kir conductance g_{Kir} and its conversion ratio α_t/α_s (labelled α_1/α_2). The five squares beside each model give the value for reservoirs of 5–25 mM ($\epsilon B = 0.5$–$2.5\,\mathrm{mM/s}$) after a 30-s input at $\epsilon = 0.1\,\mathrm{s}^{-1}$, cumulative formulation; they annotate the models and are not additional parameter positions. Asterisks mark the jointly supported models.

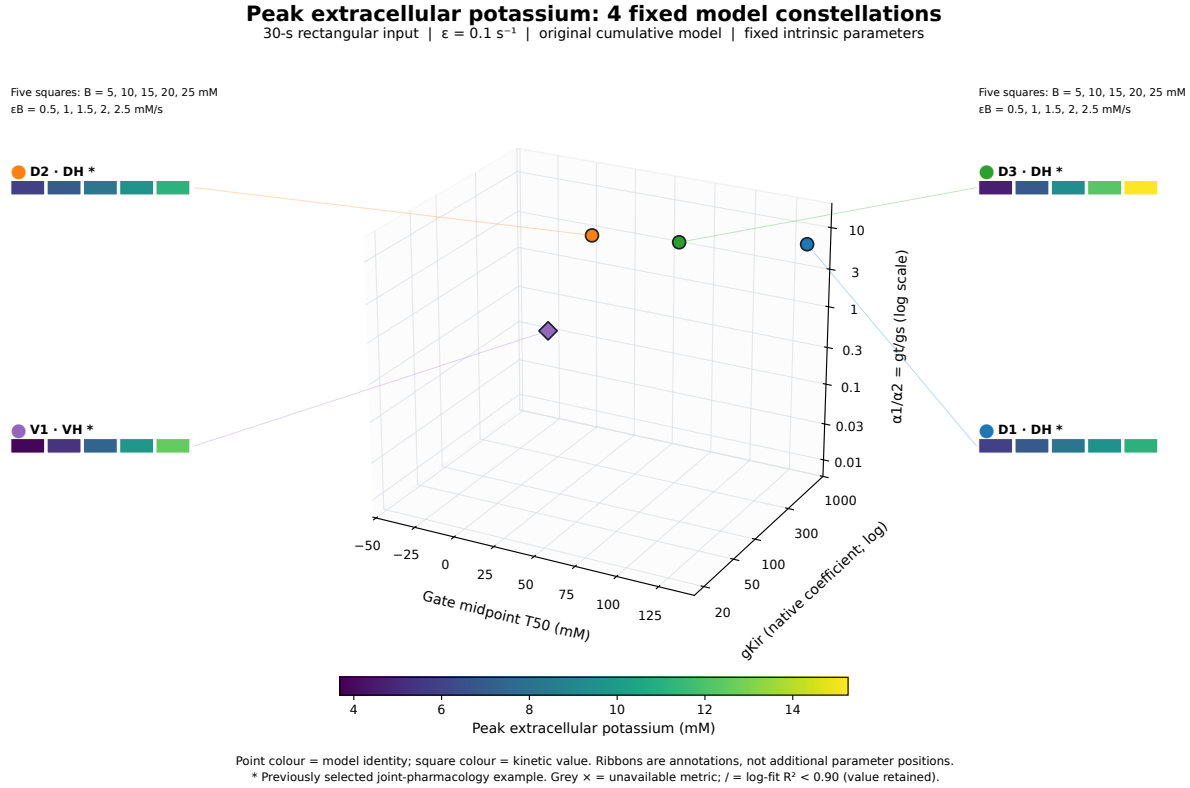


Figure D7: Peak extracellular potassium for the four jointly supported models in parameter space. As in Fig. D6, with the peak concentration.

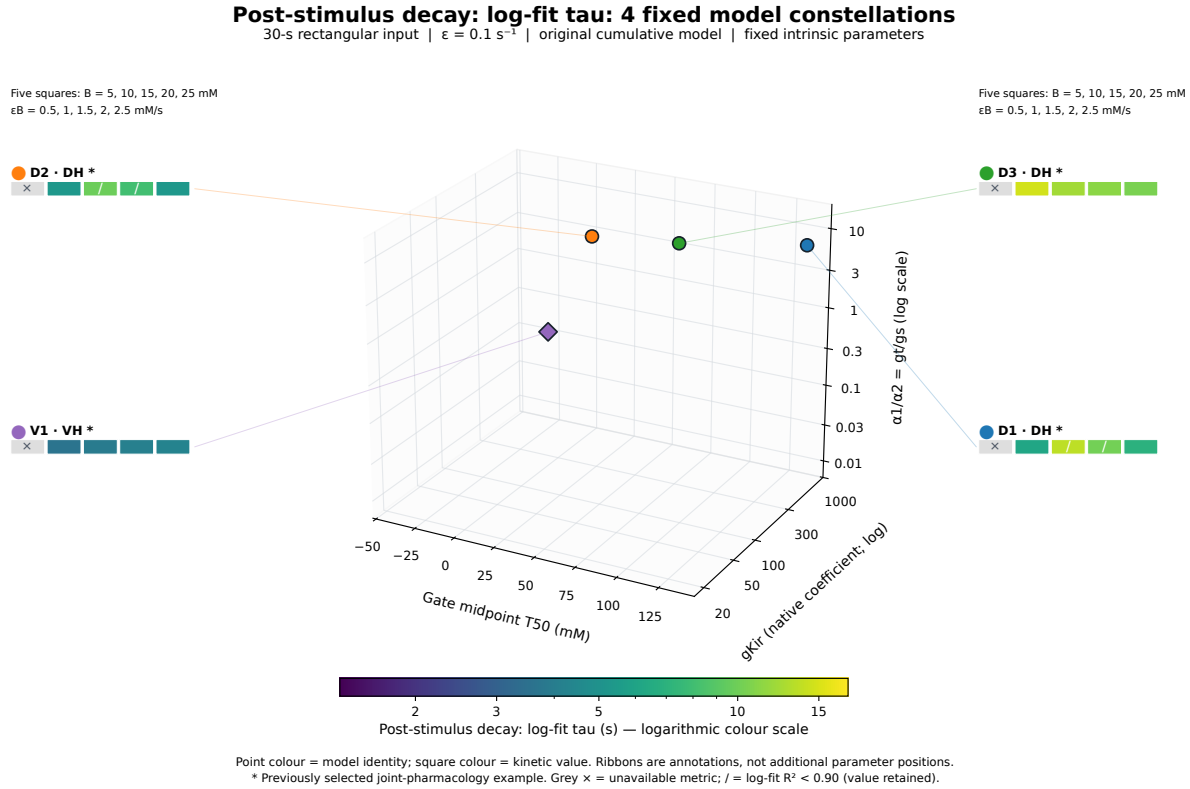


Figure D8: Decay time constant for the four jointly supported models in parameter space. As in Fig. D6, with the time constant of a log-linear fit to the post-input decay (logarithmic colour scale). Grey crosses mark loads without a valid fit interval; a slash marks a fit with $R^2 < 0.90$, whose value is retained.

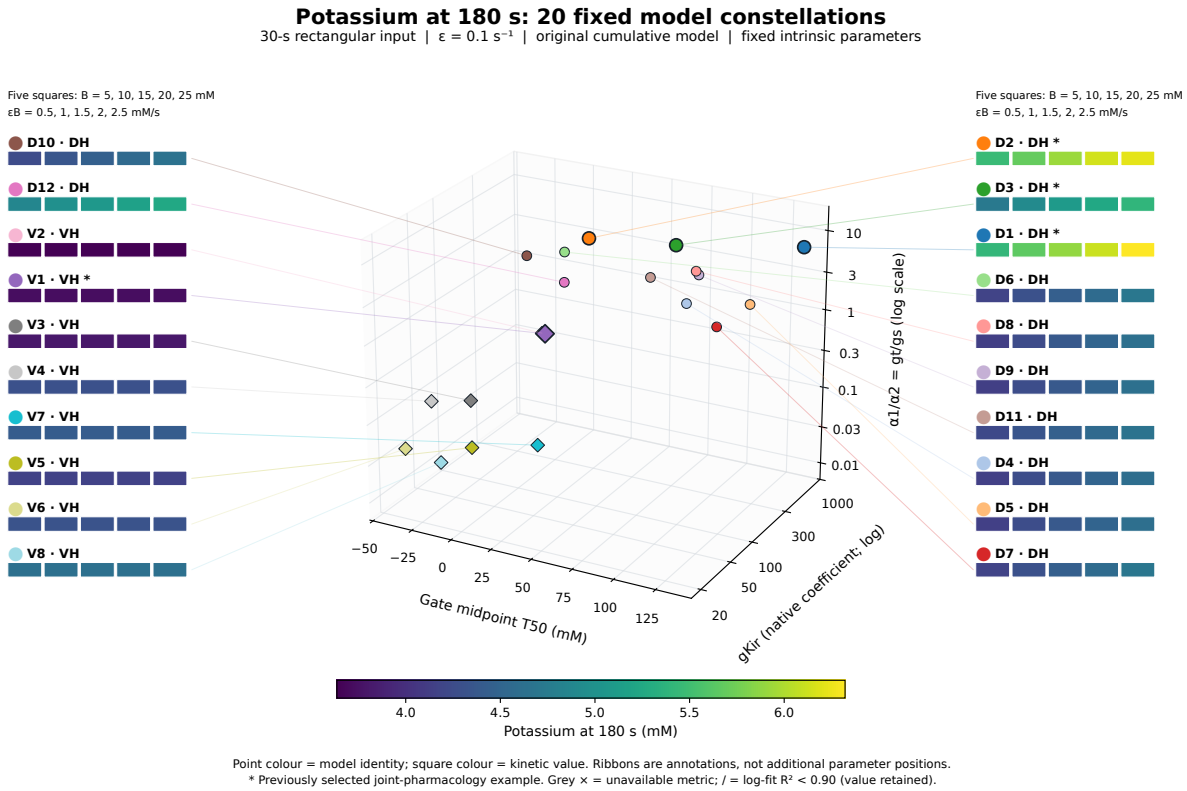


Figure D9: Potassium at 180 s for the twenty models in parameter space. As in Fig. D6, for all twenty models.

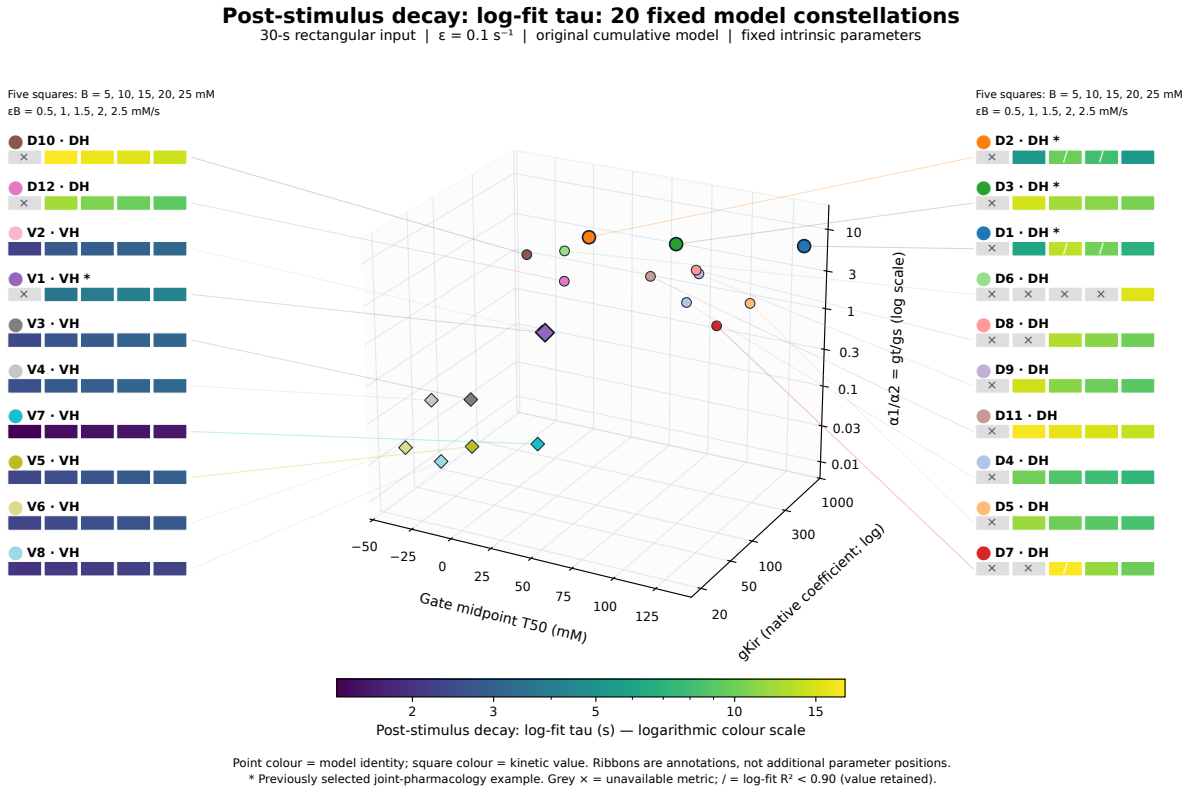


Figure D10: Decay time constant for the twenty models in parameter space. As in Fig. D8, for all twenty models.

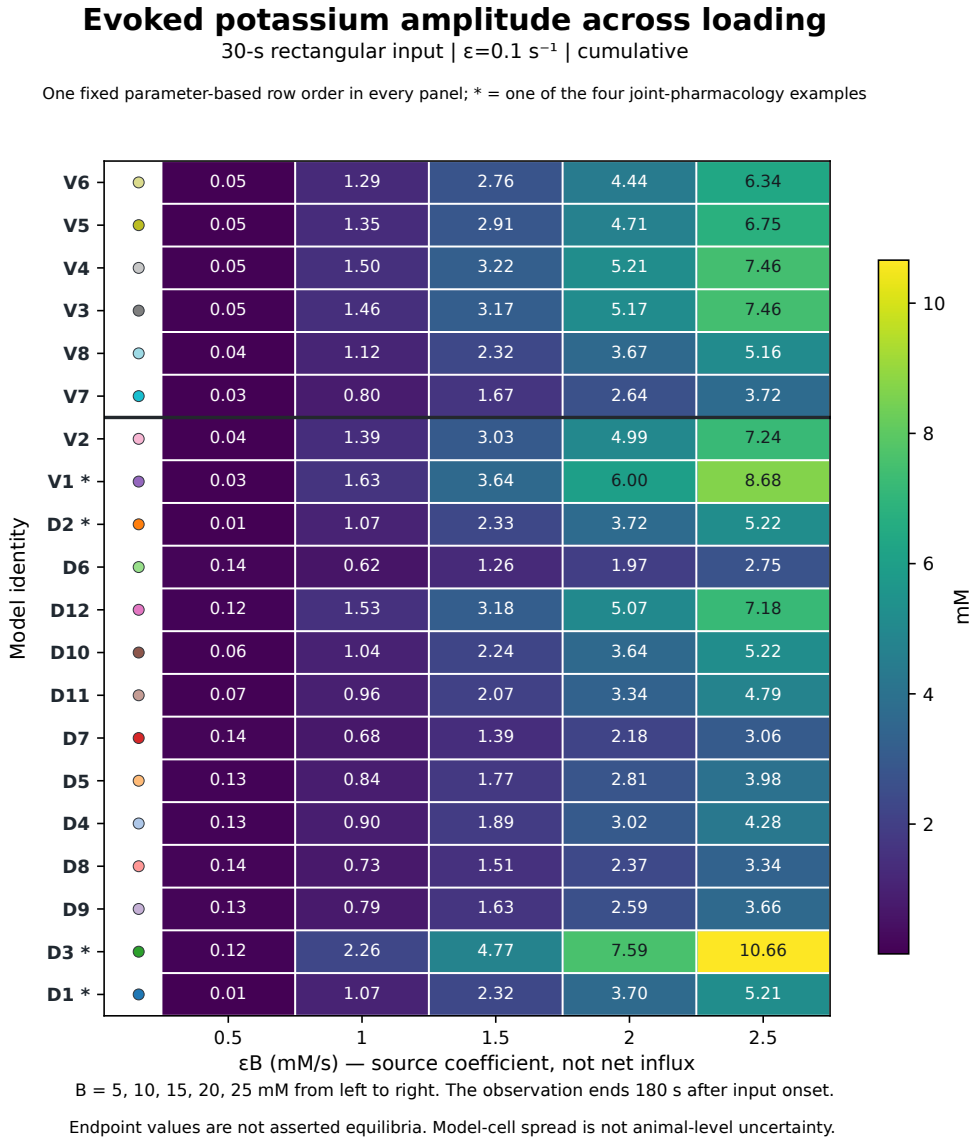


Figure D11: Evoked potassium amplitude of the twenty models across load. 30-s input at $\epsilon = 0.1\,\mathrm{s}^{-1}$, cumulative formulation. Reservoirs of 5, 10, 15, 20 and 25 mM from left to right; ϵB is the supply coefficient, not the net influx. Rows keep one parameter-based order in every map, and asterisks mark the four jointly supported models. Values at 180 s are not assumed to be stationary, and the spread among models is not animal-level uncertainty.

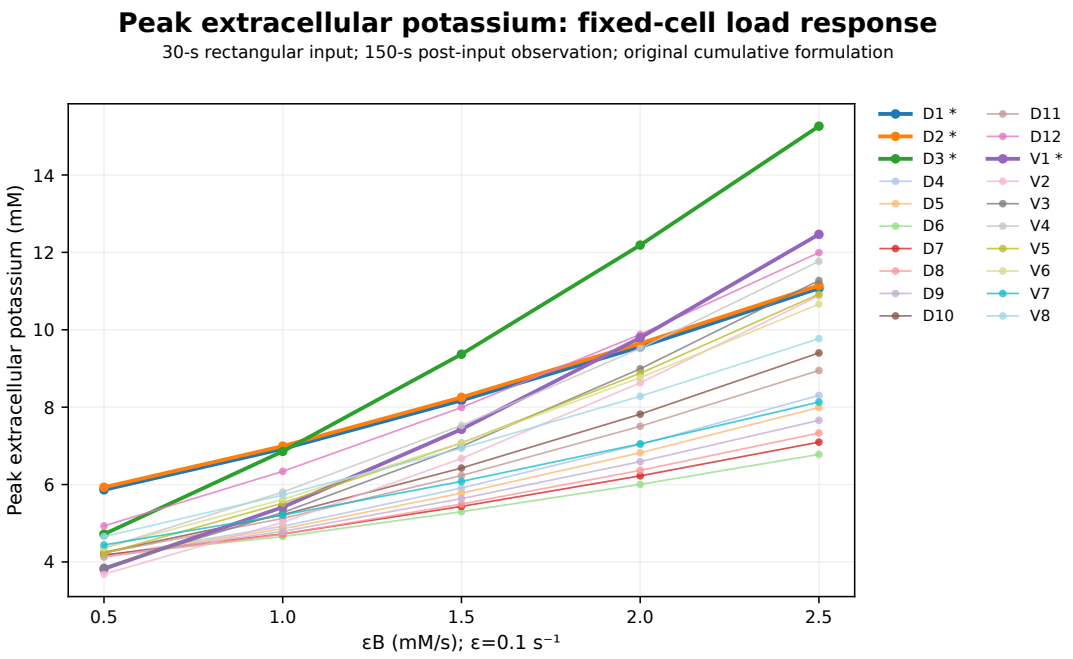


Figure D12: Peak extracellular potassium against load for the twenty models. 30-s input at $\epsilon = 0.1\,\mathrm{s}^{-1}$ and 150 s of observation after the input, cumulative formulation. Lines connect the tested loads; they are not fitted response laws, and gaps mark unavailable values. The four jointly supported models are drawn in bold.

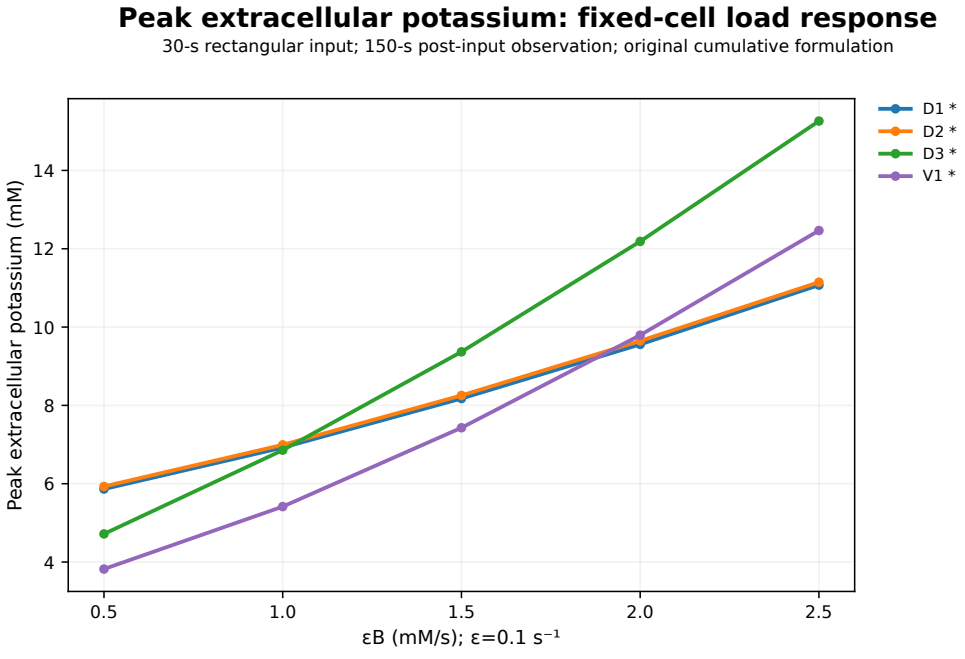


Figure D13: Peak extracellular potassium against load for the four jointly supported models. 30-s input at $\epsilon = 0.1\,\mathrm{s}^{-1}$ and 150 s of observation after the input, cumulative formulation. Lines connect the tested loads; they are not fitted response laws, and gaps mark unavailable values.

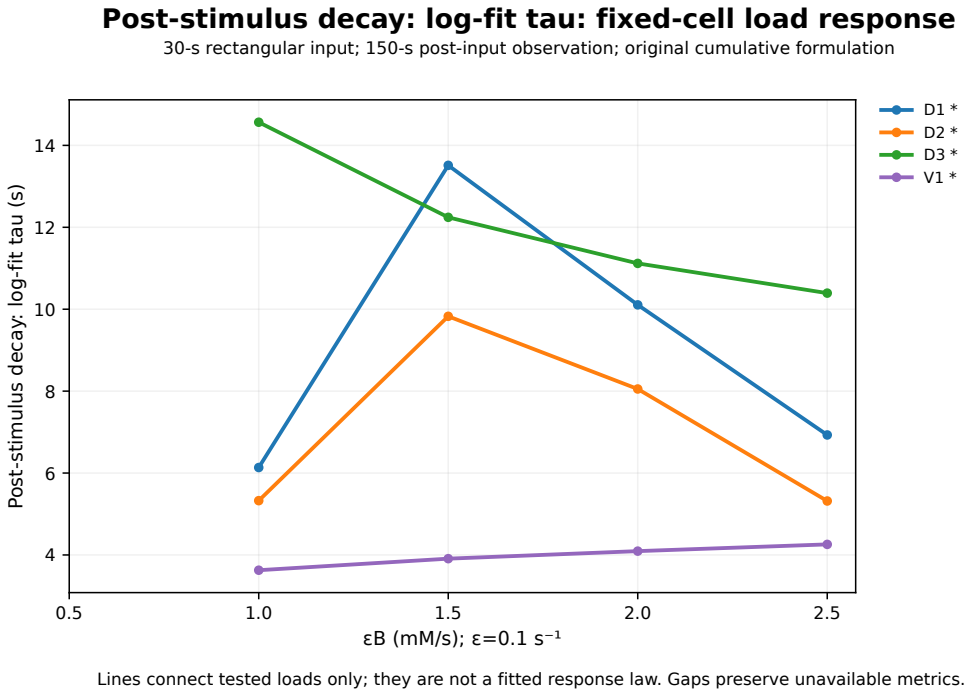


Figure D14: Decay time constant against load for the four jointly supported models. 30-s input at $\epsilon = 0.1\,\mathrm{s}^{-1}$ and 150 s of observation after the input, cumulative formulation. Lines connect the tested loads; they are not fitted response laws, and gaps mark unavailable values. No valid fit interval exists at the lowest load.

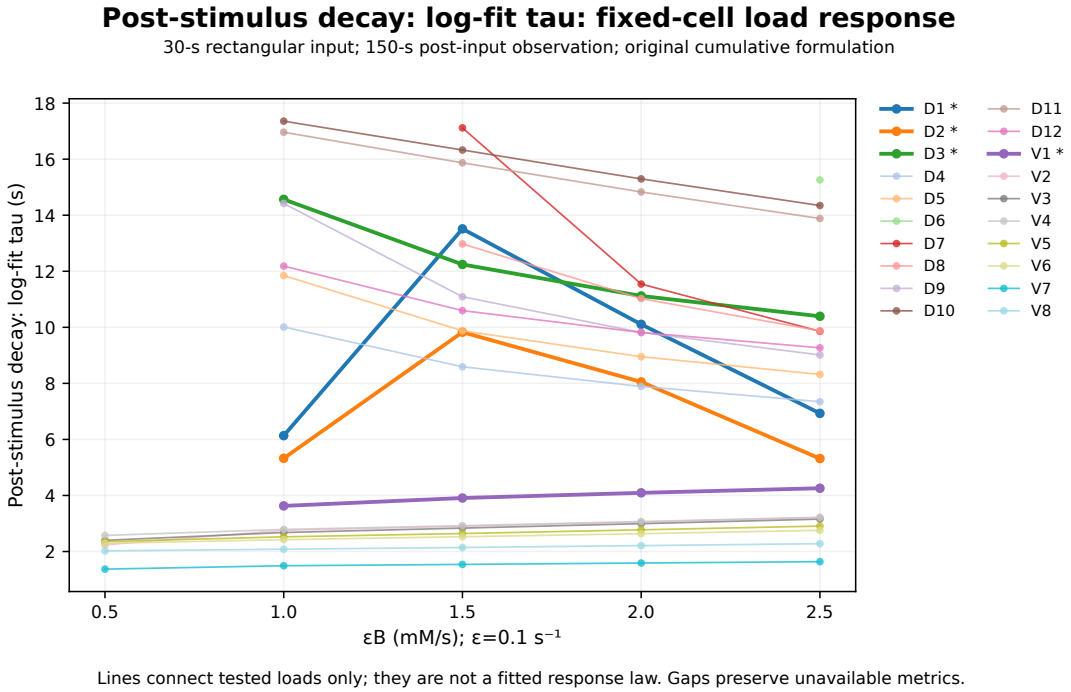


Figure D15: Decay time constant against load for the twenty models. 30-s input at $\epsilon = 0.1\,\mathrm{s}^{-1}$ and 150 s of observation after the input, cumulative formulation. Lines connect the tested loads; they are not fitted response laws, and gaps mark unavailable values. The four jointly supported models are drawn in bold.

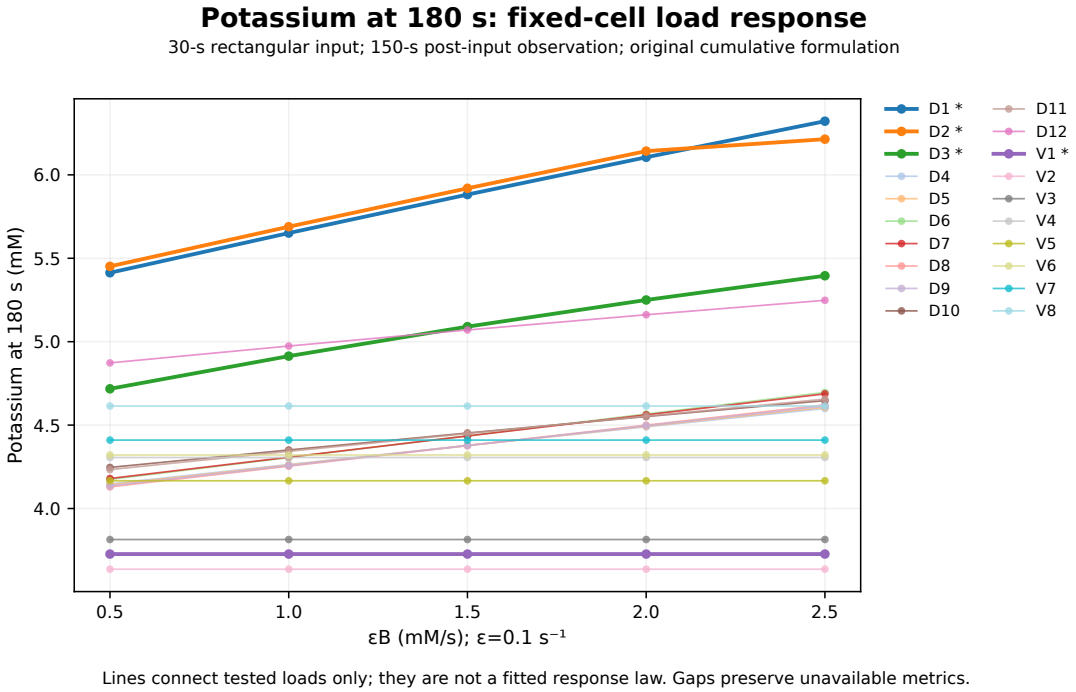


Figure D16: Potassium at 180 s against load for the twenty models. 30-s input at $\epsilon = 0.1\,\mathrm{s}^{-1}$ and 150 s of observation after the input, cumulative formulation. Lines connect the tested loads; they are not fitted response laws, and gaps mark unavailable values. The four jointly supported models are drawn in bold.

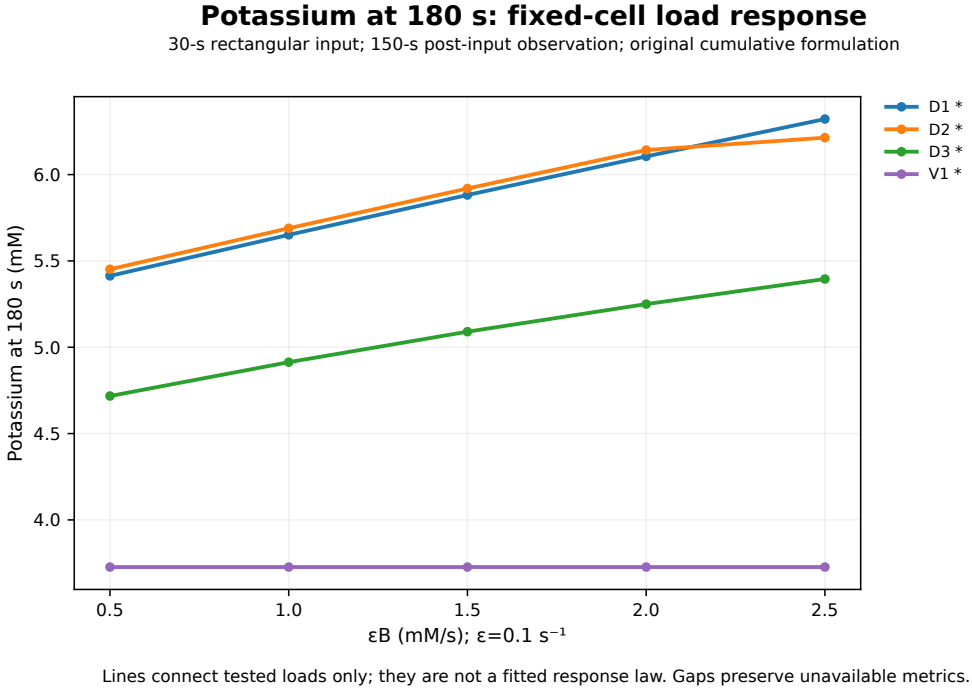


Figure D17: Potassium at 180 s against load for the four jointly supported models. 30-s input at $\epsilon = 0.1\,\mathrm{s}^{-1}$ and 150 s of observation after the input, cumulative formulation. Lines connect the tested loads; they are not fitted response laws, and gaps mark unavailable values.

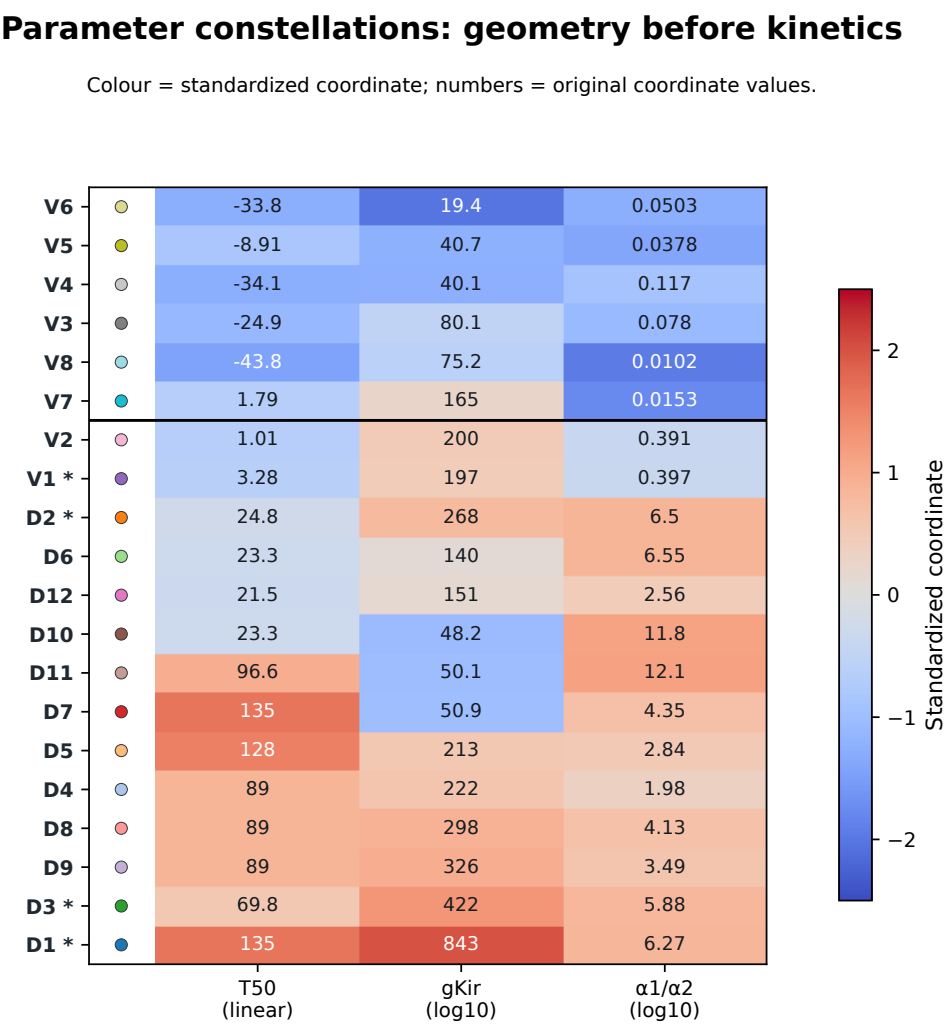


Figure D18: Parameter coordinates of the twenty models. Gate midpoint T_{50} (linear), $\log_{10} g_{\mathrm{Kir}}$ and $\log_{10}(\alpha_t/\alpha_s)$; colour shows the standardized coordinate and numbers the original values. Rows are ordered by Ward clustering (Ward, 1963) of these three coordinates only, without regional labels or potassium responses. The coordinates are projections of complete parameter sets, in which coupling, absolute conversion, geometry and leak also vary.

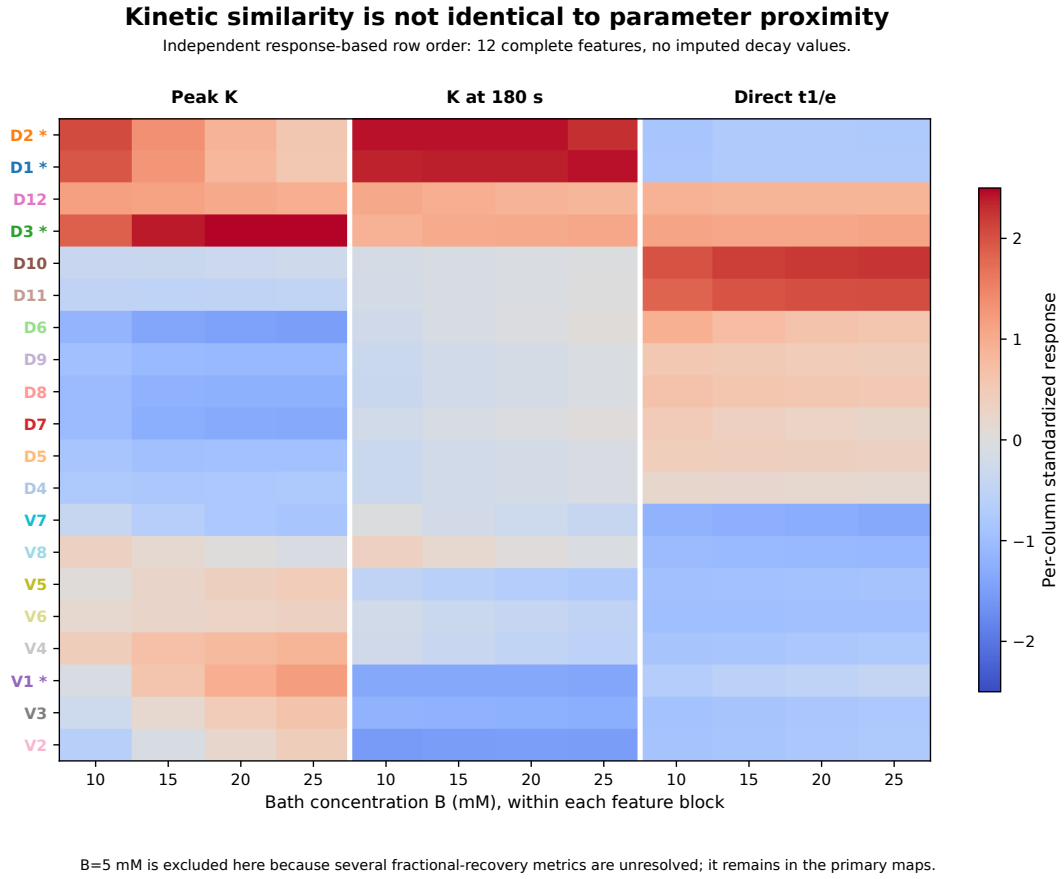


Figure D19: Response similarity does not follow parameter proximity. Standardized peak potassium, potassium at 180 s and direct $t_{1/e}$ for reservoirs of 10–25 mM (twelve complete features, with no imputed decay values); rows are ordered by response. The 5-mM level is excluded because several recovery descriptors are unavailable there; it remains in the load maps.

Recruitment comparisons at the lower reservoir



Figure D20: Paired benefit of recruitment. Positive values indicate less extracellular accumulation when the same pathway is allowed to recruit dynamically. The benefit is not a global maximum of capacity. K_{frozen} on the panel denotes the concentration with recruitment held.



Figure D21: Signed flux balance in D1. The fluxes are model quantities. A net extracellular decline requires βJ_t to exceed replenishment from the reservoir.

Coupling loss with Kir retained

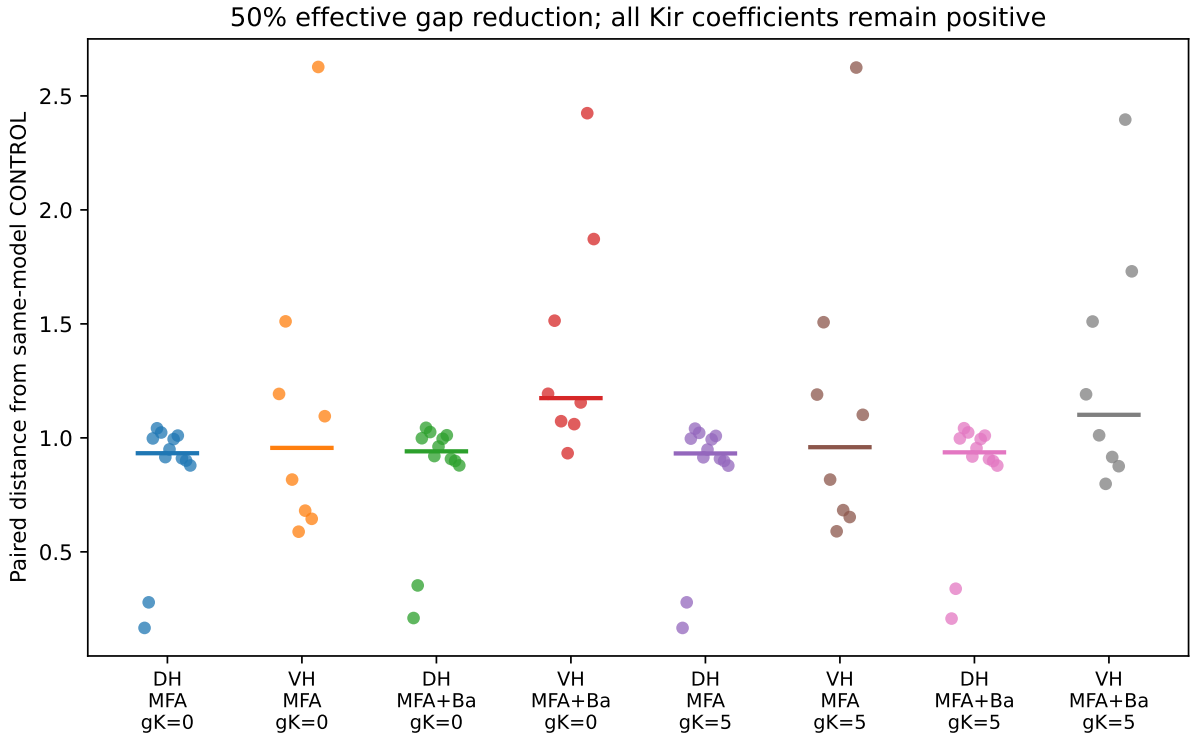


Figure D22: Paired displacement of recovery profiles after coupling loss. Distance of each model from its own CONTROL profile after MFA-like coupling loss, alone or combined with partial Kir reduction (MFA+Ba), for $g_{K,a} = 0$ and $g_{K,a} = 5$. Dots are individual models and horizontal marks medians. Each variant is measured in the geometry of its own CONTROL responses, so distance magnitudes are not a common physical unit across variants. Five-feature kinetic profile at 10–25 mM; recoveries unavailable at low load are not set to zero.

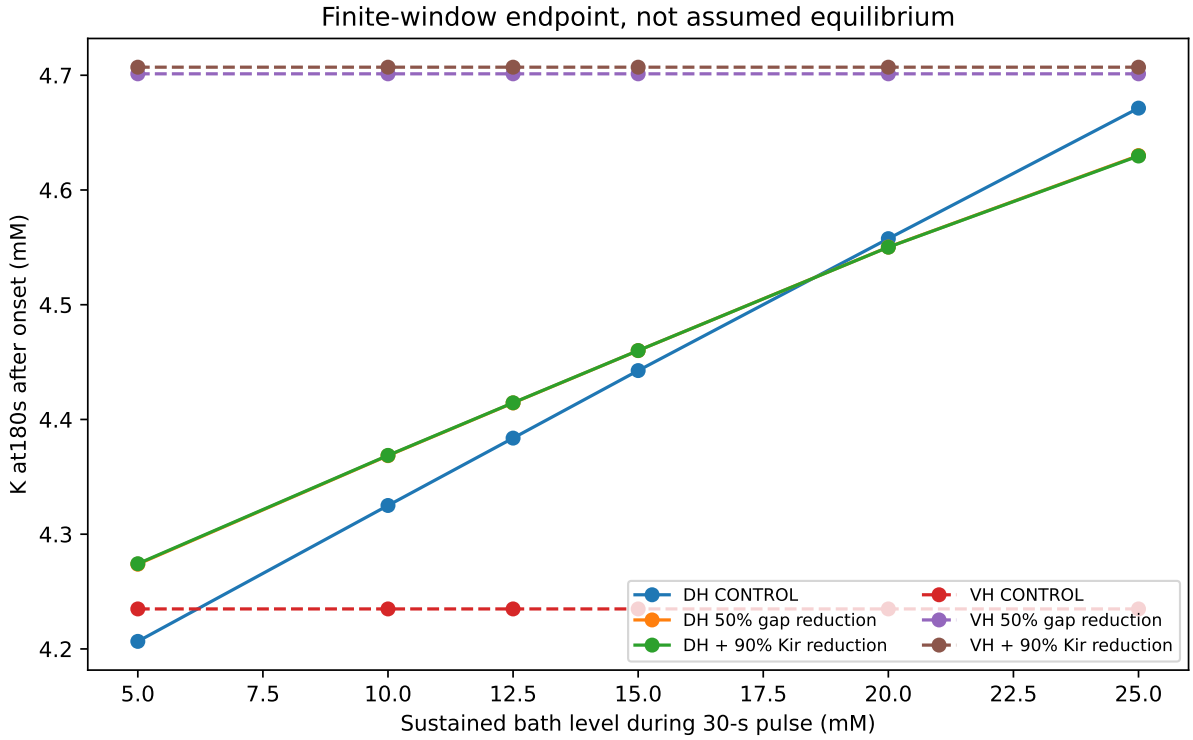


Figure D23: Potassium at the end of the finite window, which is not assumed to be an equilibrium. Regional medians across models, not animal confidence intervals. The reservoir waveform and exchange are the same in every condition; $g_{K,a} = 5$. The combined intervention retains 10% of the Kir conductance. Missing recoveries remain absent.

Earlier profile-robustness analysis at a different intervention scale

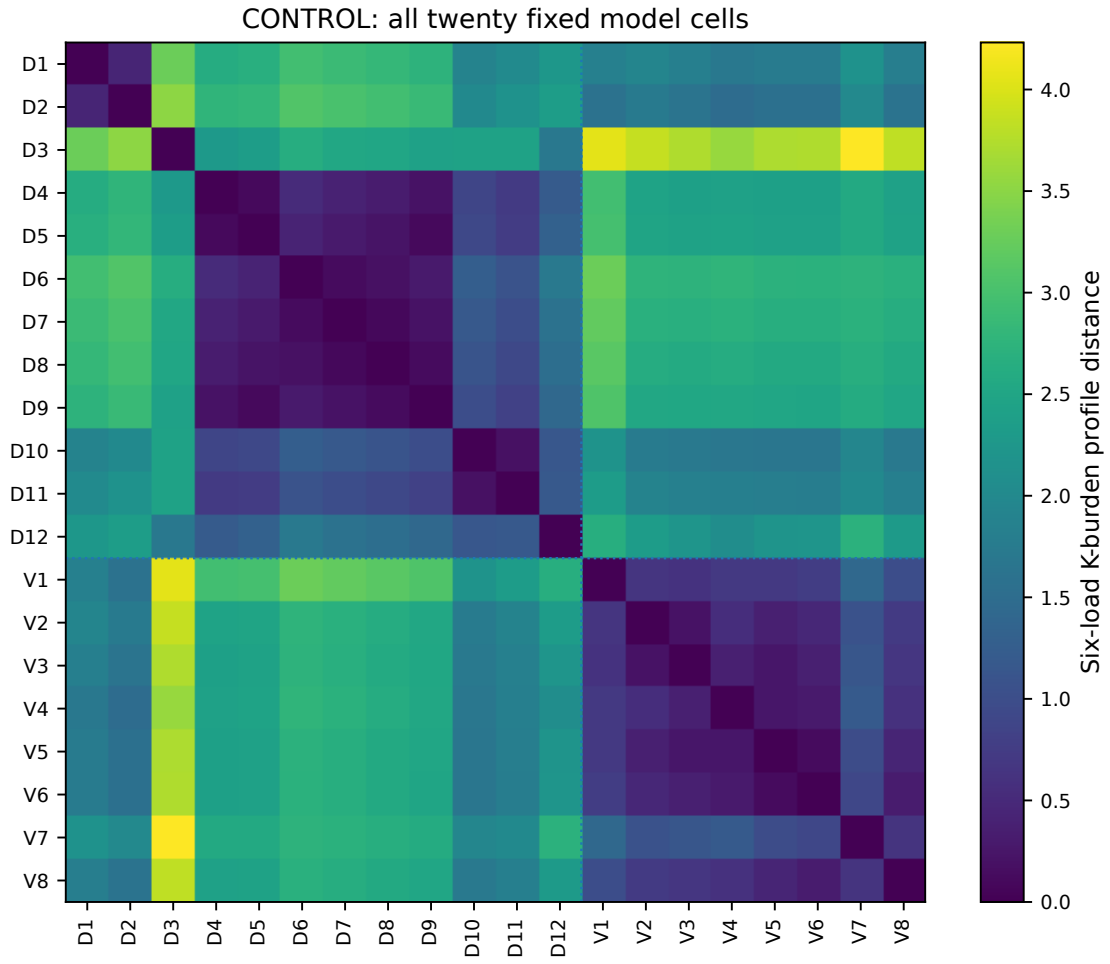


Figure D24: Whole-cell baseline potassium geometry. Earlier analysis. Distances between models computed from three descriptors available for every response: log absolute peak, transformed positive exposure and signed residual. The same CONTROL-defined metric is used for both groups and for every intervention. Lines separate the twelve DH-like and eight VH-like models. The values are distances between model profiles, not probabilities of biological acceptance.

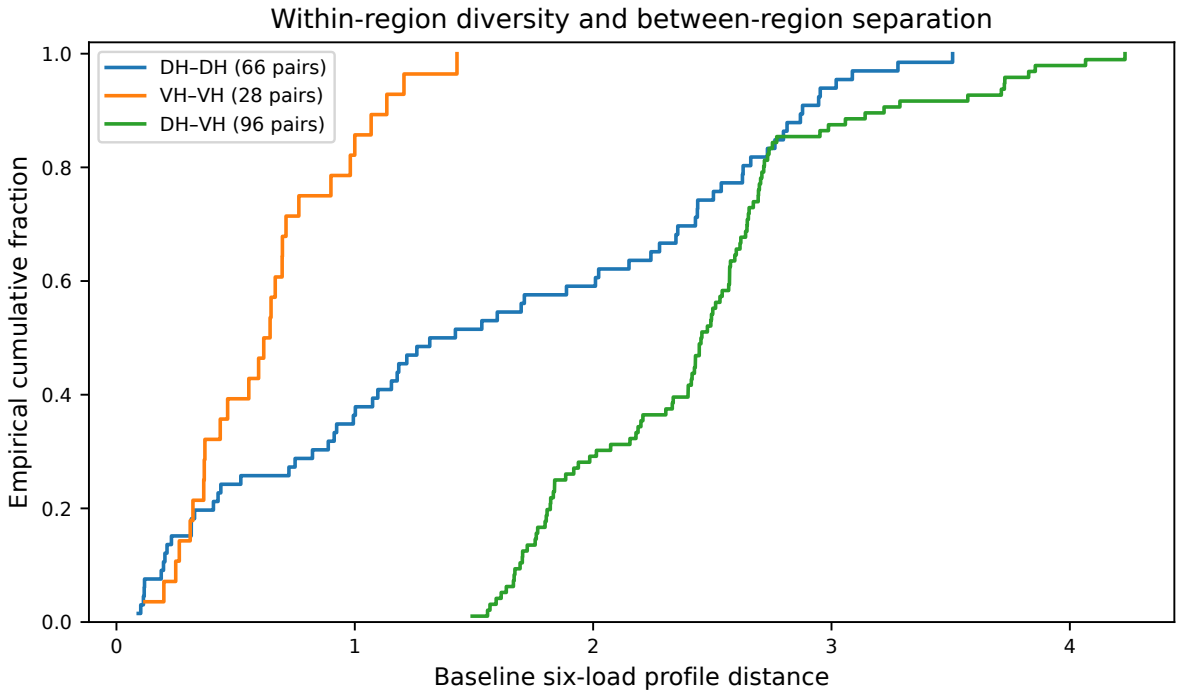


Figure D25: Distributions of baseline distances. Earlier analysis. Pairs share models and are not independent samples. Because comparing the two within-group distributions alone would miss a shift between groups, the cross-region distribution is also shown. No p value is computed from pairs.

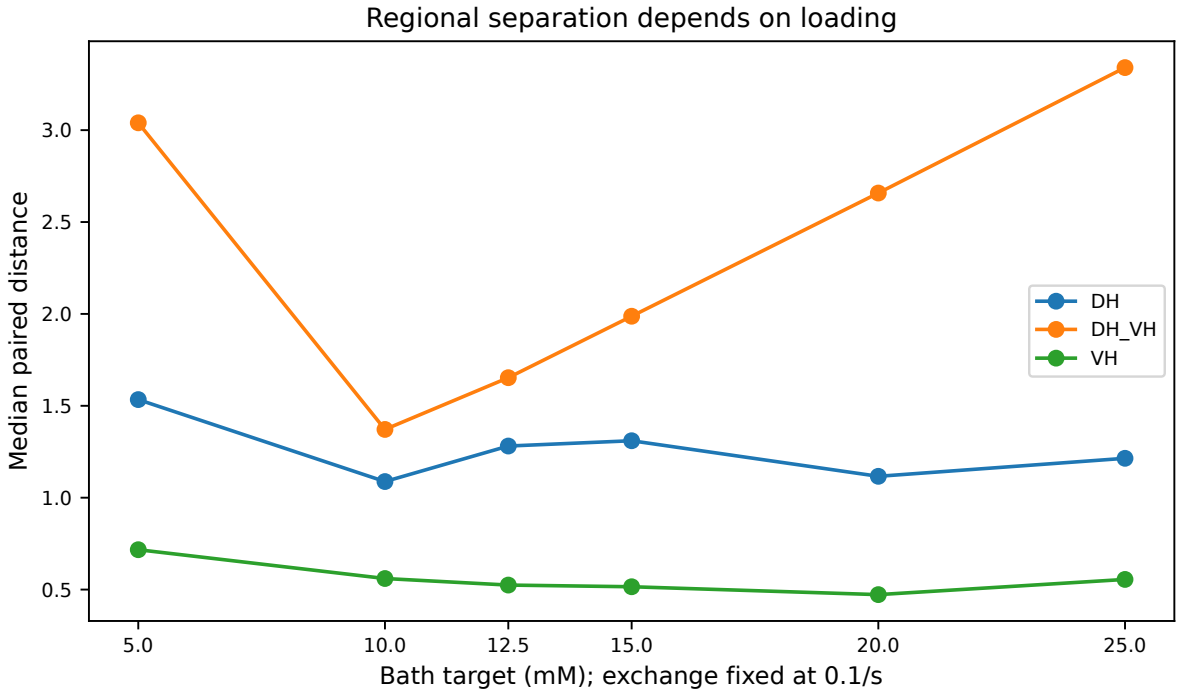


Figure D26: Distance structure at each load. Earlier analysis. The coordinate scale is the same at every load, and the median pairwise distance is descriptive. Matching the product of exchange coefficient and reservoir concentration would not ensure matching washout, so both are specified.

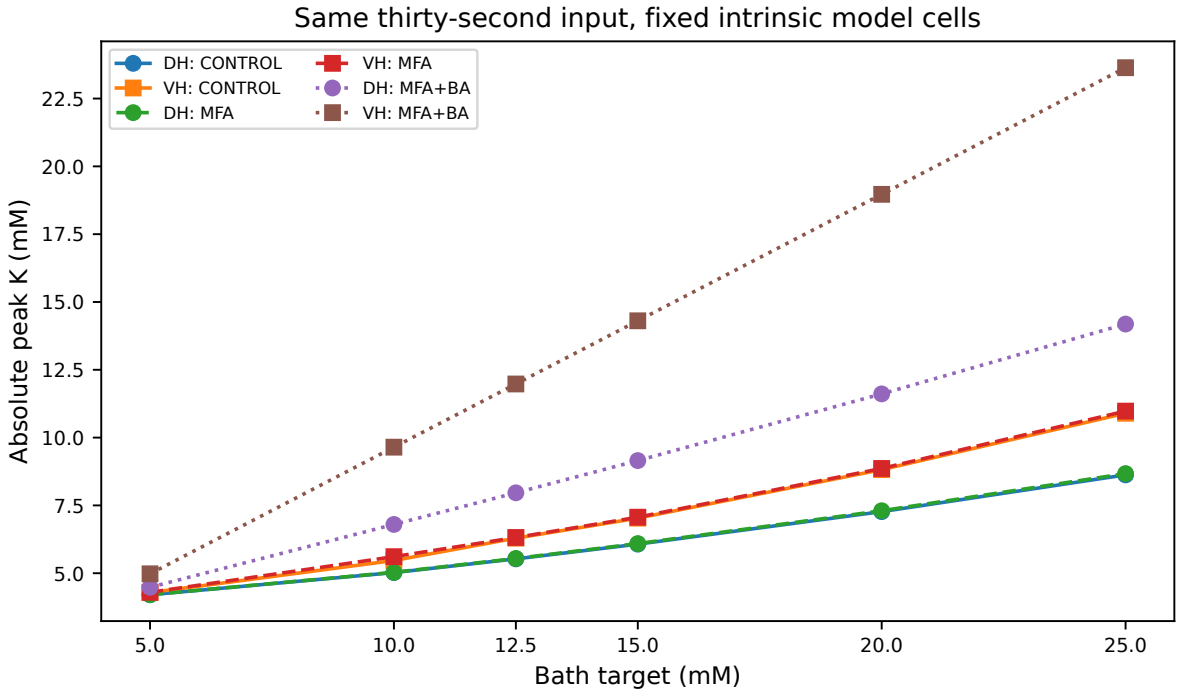


Figure D27: Absolute peak potassium across load. Earlier analysis. Lines summarize the model groups at each load; the number of valid recoveries per load is shown in Fig. D31. Missing crossings are not plotted as zero. These are summaries of the model ensemble, not animal confidence intervals.

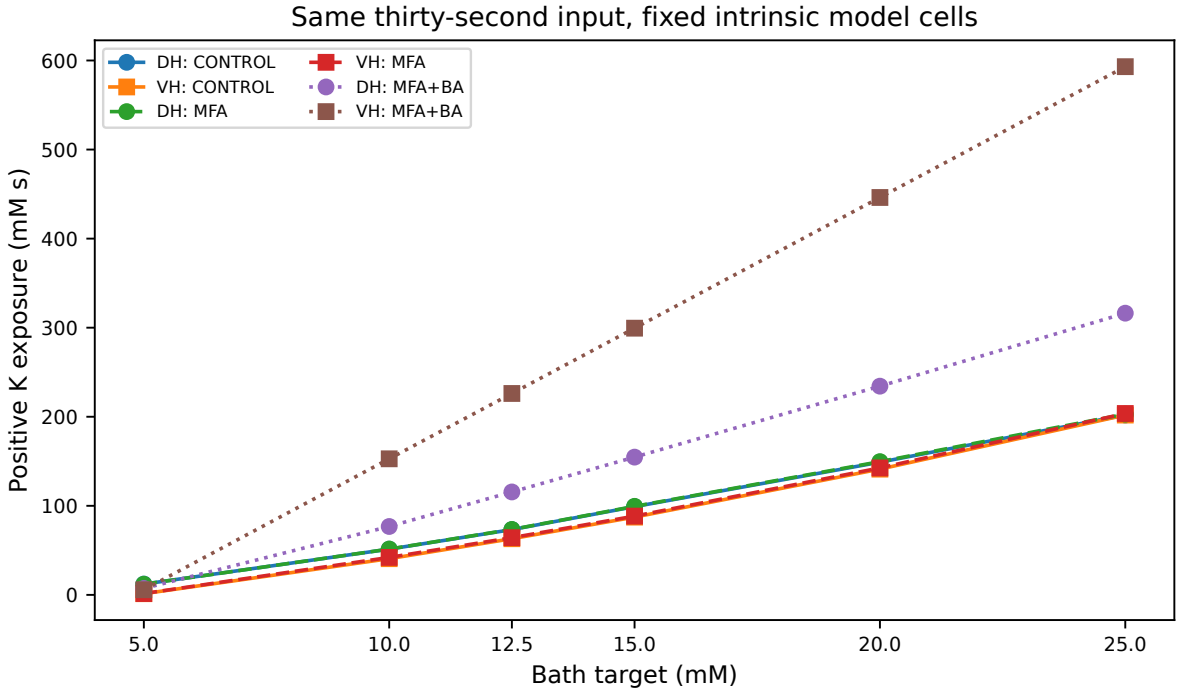


Figure D28: Positive potassium exposure across load. Earlier analysis. Lines summarize the model groups at each load; the number of valid recoveries per load is shown in Fig. D31. Missing crossings are not plotted as zero. These are summaries of the model ensemble, not animal confidence intervals.

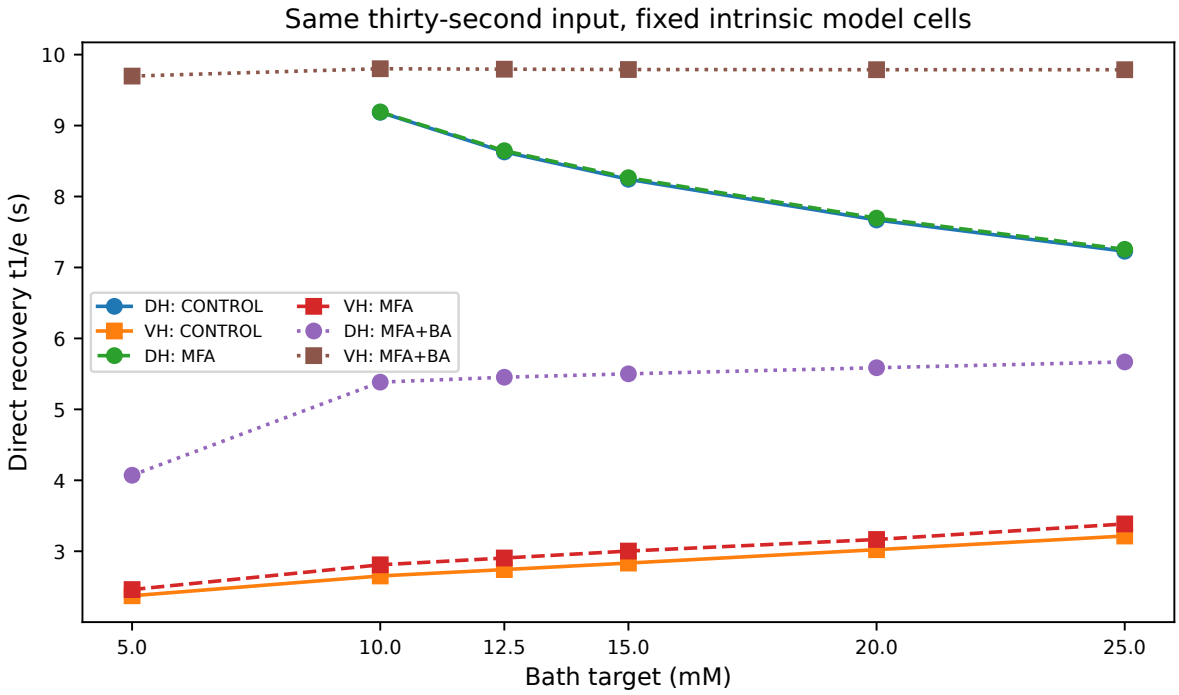


Figure D29: Direct recovery time $t_{1/e}$ across load. Earlier analysis. Lines summarize the model groups at each load; the number of valid recoveries per load is shown in Fig. D31. Missing crossings are not plotted as zero. These are summaries of the model ensemble, not animal confidence intervals.

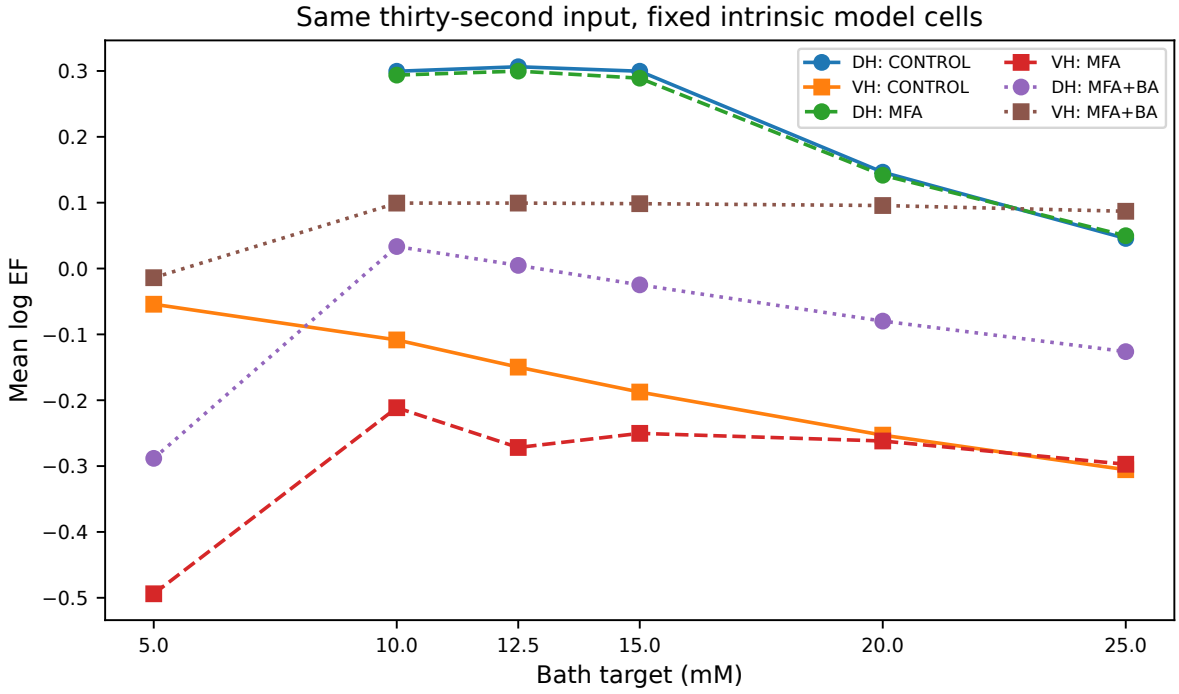


Figure D30: Mean log EF across load. Earlier analysis. Lines summarize the model groups at each load; the number of valid recoveries per load is shown in Fig. D31. Missing crossings are not plotted as zero. These are summaries of the model ensemble, not animal confidence intervals. The mean of $\log EF$ corresponds to a geometric summary of EF. A flatter EF does not necessarily mean a smaller peak, a smaller exposure or a faster absolute recovery.

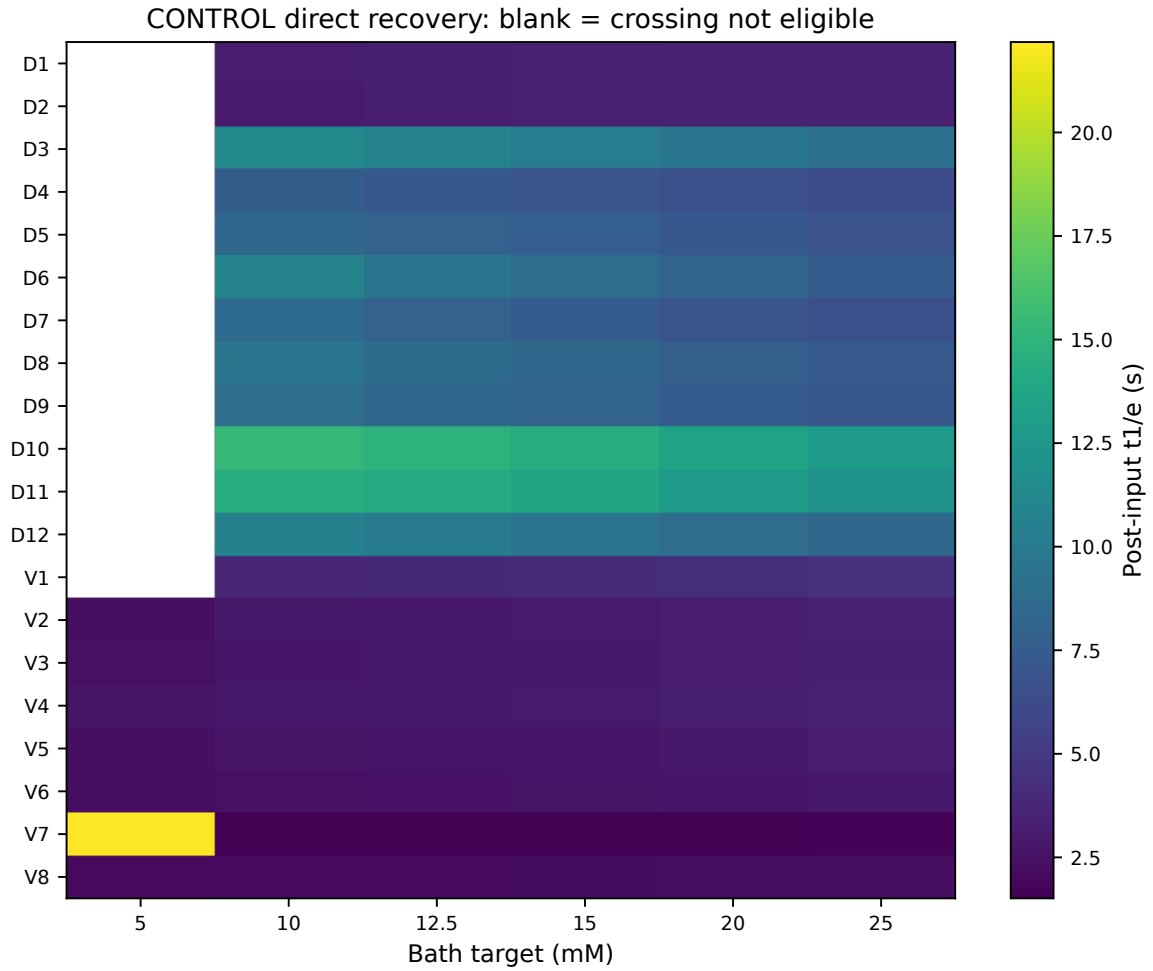


Figure D31: Availability of recovery at each load. Earlier analysis. No DH-like model has an valid post-input $t_{1/e}$ at the lowest (5-mM) reservoir. Some responses cross before the input ends or show relaxation of the baseline instead of an evoked recovery. The burden profile keeps all six loads, whereas the prespecified recovery profile uses 10–25 mM.

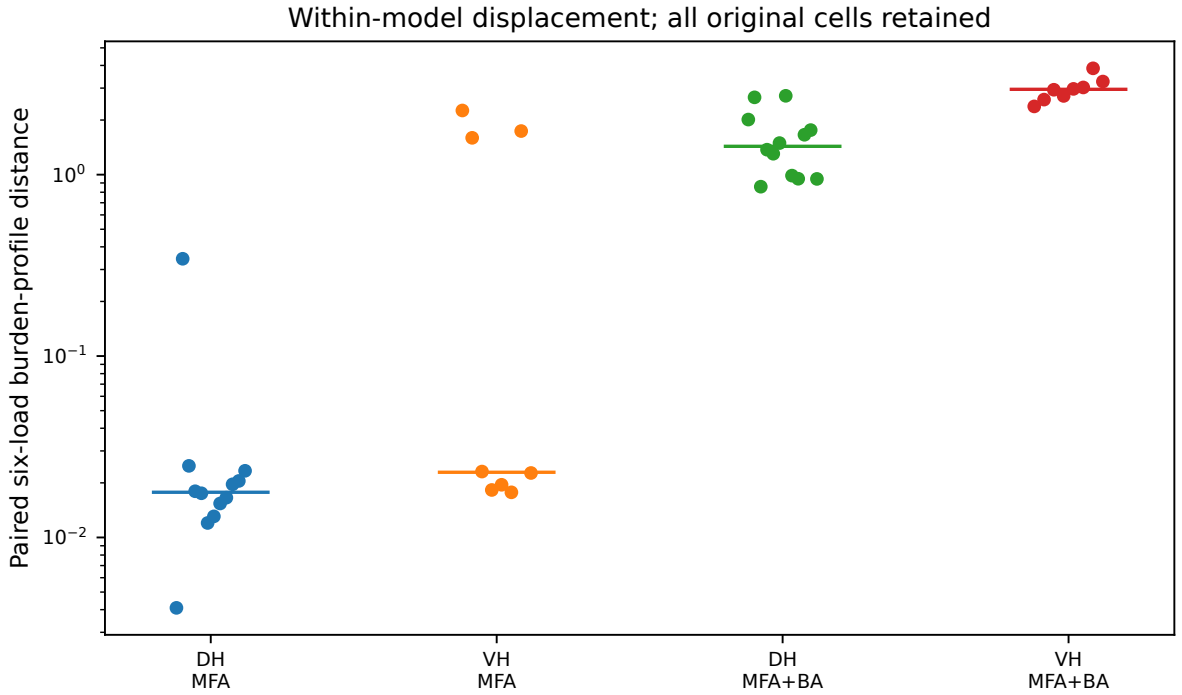


Figure D32: Displacement of the twenty-model panel under intervention. Earlier analysis. Dots are models and horizontal bars regional medians. A smaller distance means greater preservation in this feature space, not necessarily better potassium regulation. No filter based on the pharmacological results is applied.

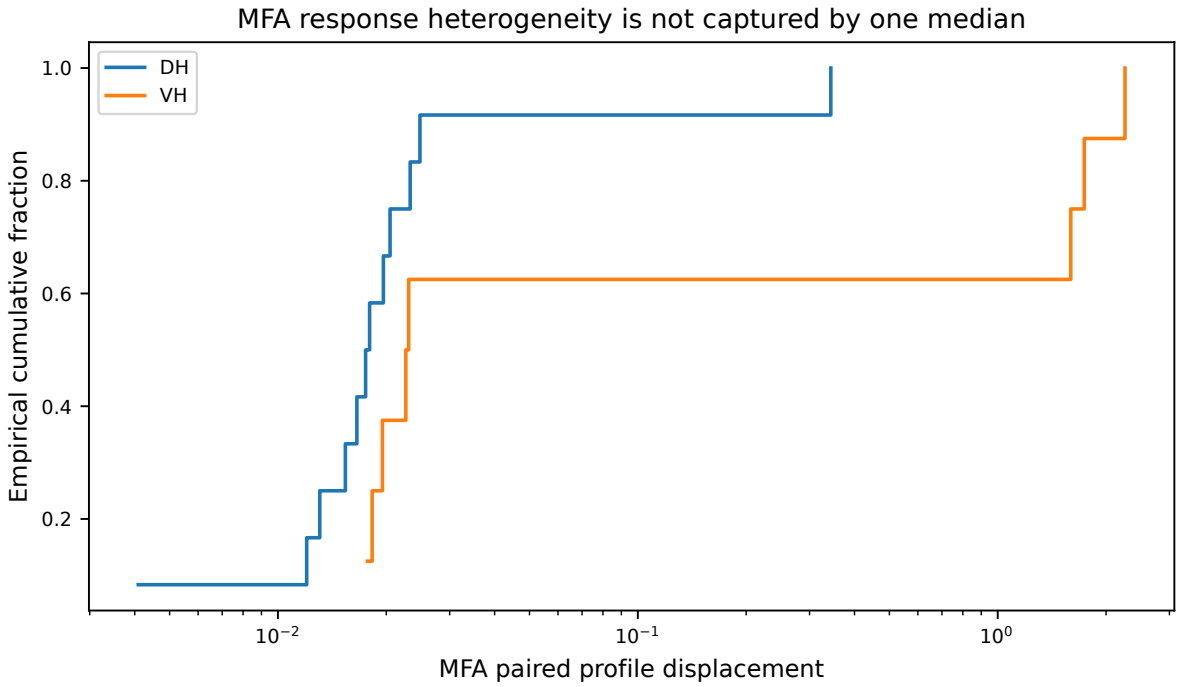


Figure D33: Displacement distributions under the MFA-like intervention. Earlier analysis. The shared MFA-like mapping has small typical effects but much larger effects in particular models, so both the median and the tail are shown. The distributions are conditional on this specified intervention.

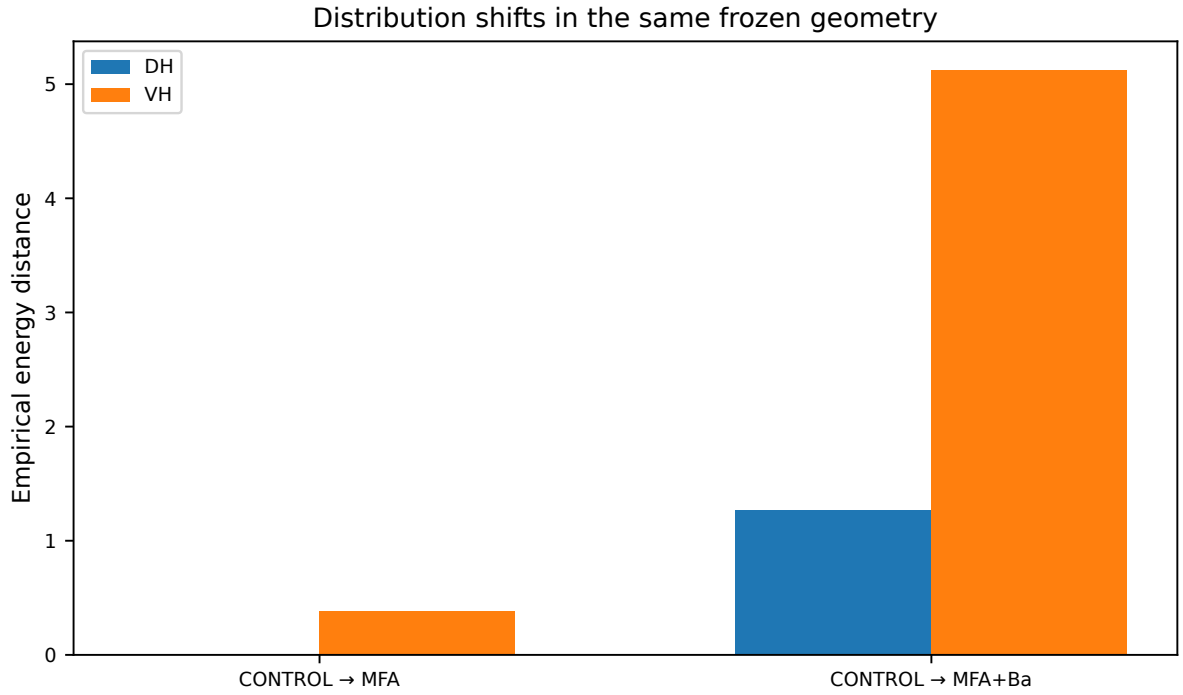


Figure D34: Shifts of the regional distributions under intervention. Earlier analysis. The energy distance (Székely and Rizzo, 2013) compares complete empirical distributions in the fixed profile representation and complements the paired displacement. The same models contribute before and after intervention. "Frozen geometry on the panel denotes this representation, fixed before intervention.

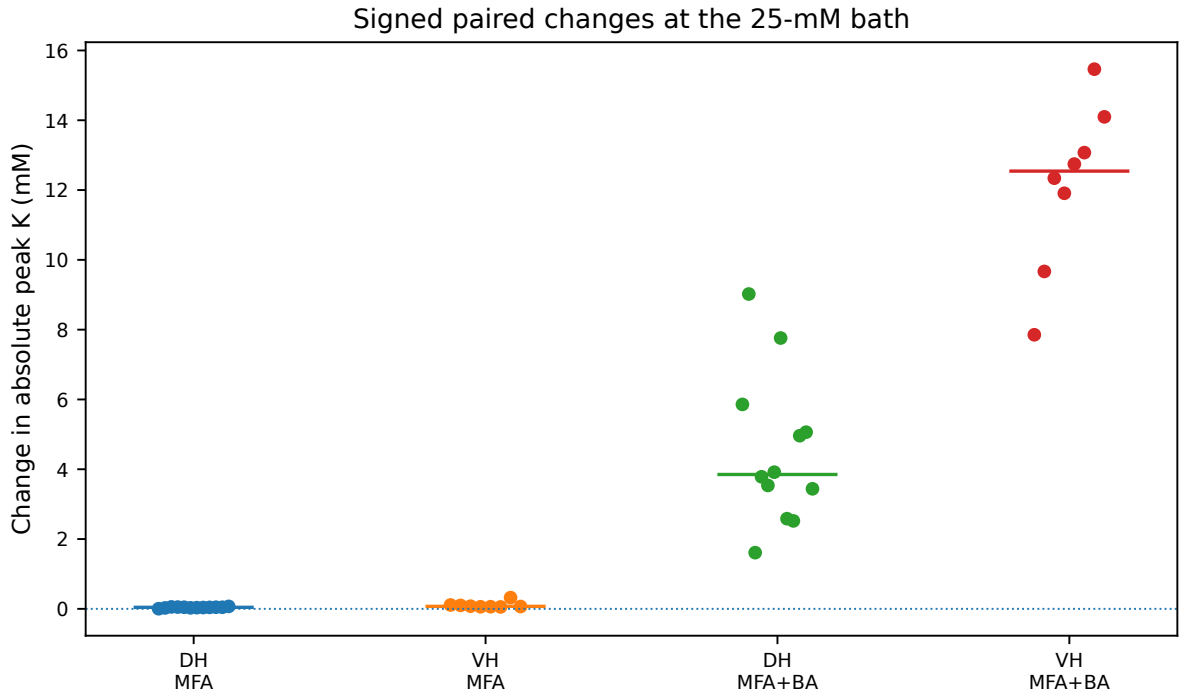


Figure D35: Change in absolute peak potassium. Earlier analysis. Each dot compares a model with its own CONTROL response under identical external loading. The sign matters: a change in recovery time can differ in sign from the change in peak or exposure.

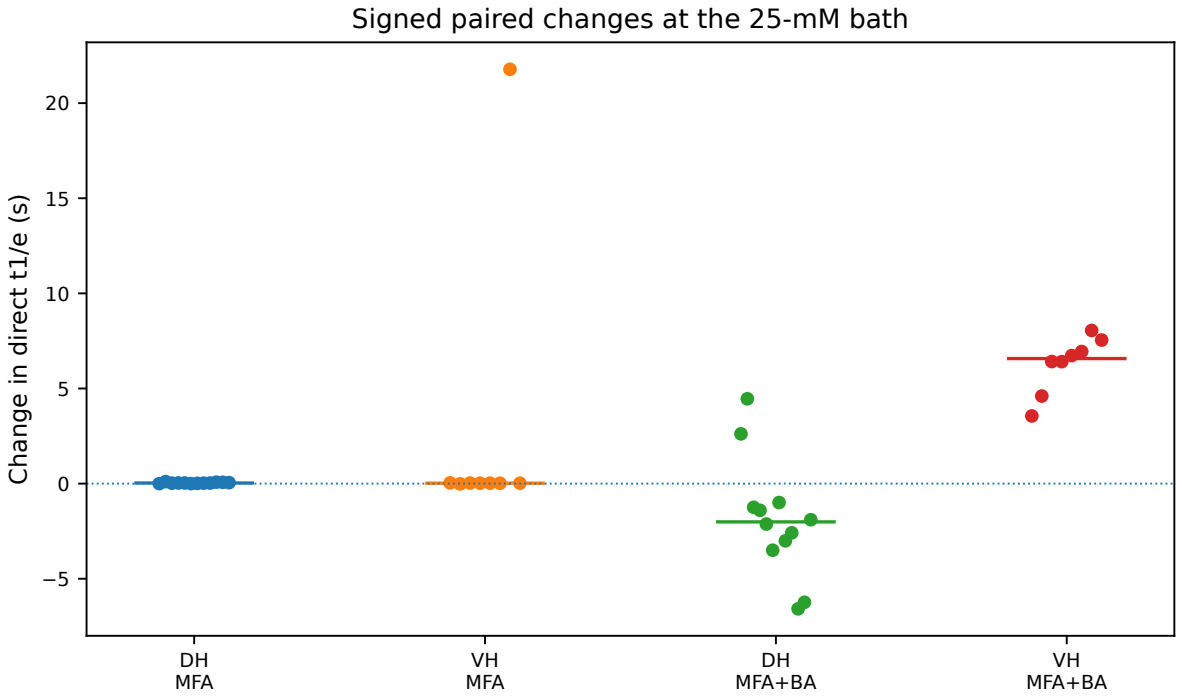


Figure D36: Change in direct $t_{1/e}$. Earlier analysis. Each dot compares a model with its own CONTROL response under identical external loading. The sign matters: a change in recovery time can differ in sign from the change in peak or exposure.

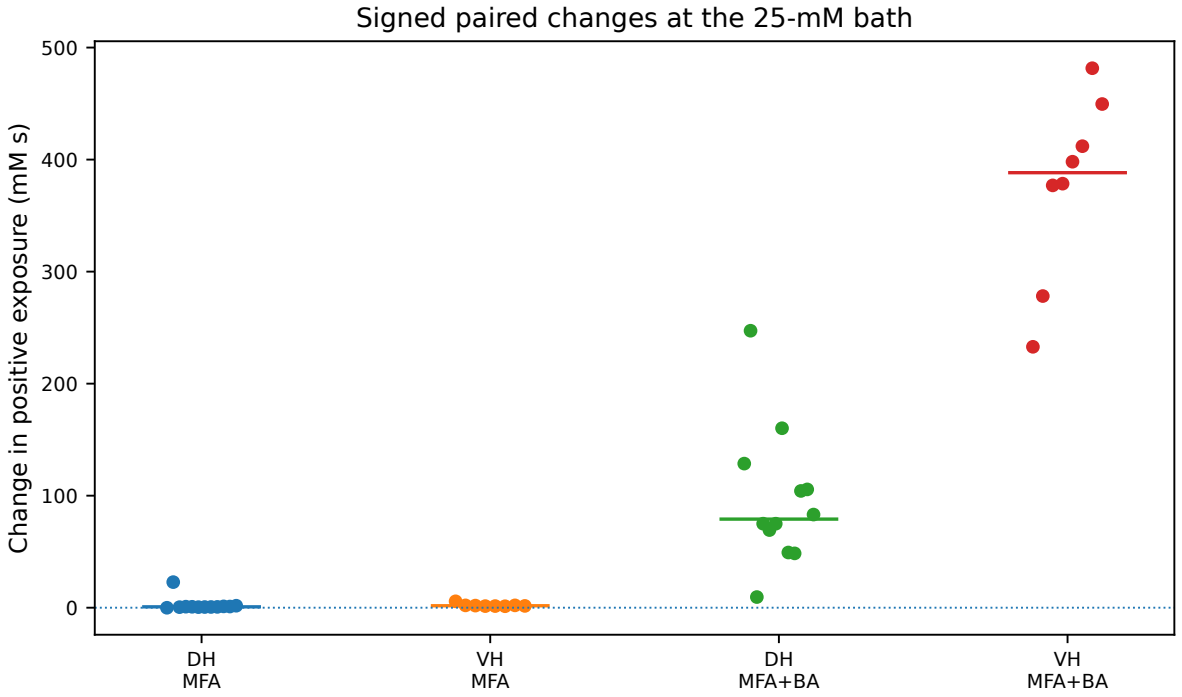


Figure D37: Change in positive exposure. Earlier analysis. Each dot compares a model with its own CONTROL response under identical external loading. The sign matters: a change in recovery time can differ in sign from the change in peak or exposure.

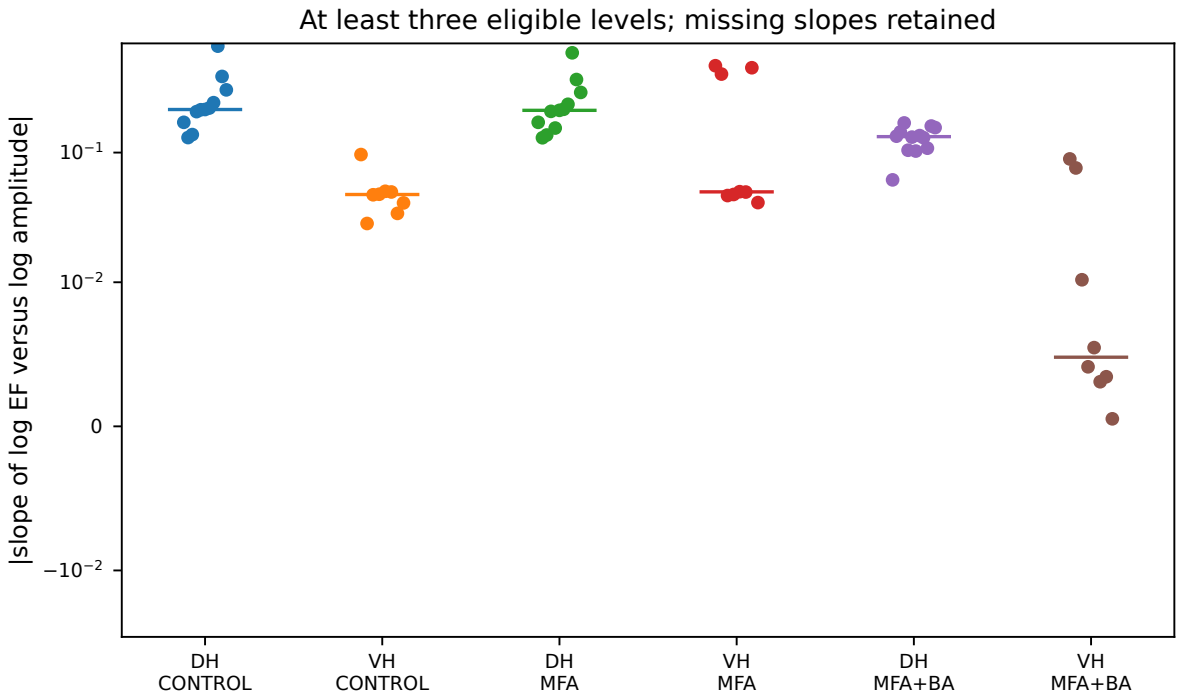


Figure D38: Absolute slope of $\log EF$ **against log amplitude.** Earlier analysis. Signed and absolute sensitivities are kept separate, and the load is varied within each model. The slope against observed amplitude corresponds to the definition used for the tissue analysis (Appendix B); the sensitivity to the imposed input has a different predictor. A flat response can coexist with a compromised mechanism.

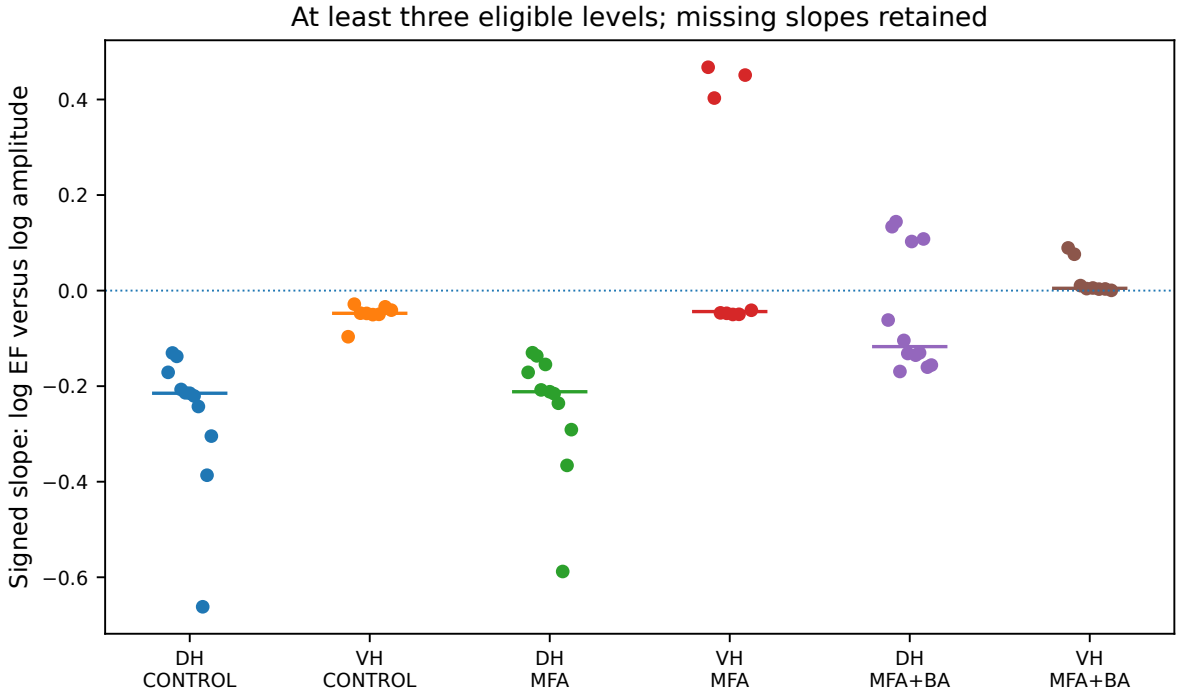


Figure D39: Signed slope of $\log EF$ **against log amplitude.** Earlier analysis. Signed and absolute sensitivities are kept separate, and the load is varied within each model. The slope against observed amplitude corresponds to the definition used for the tissue analysis (Appendix B); the sensitivity to the imposed input has a different predictor. A flat response can coexist with a compromised mechanism.

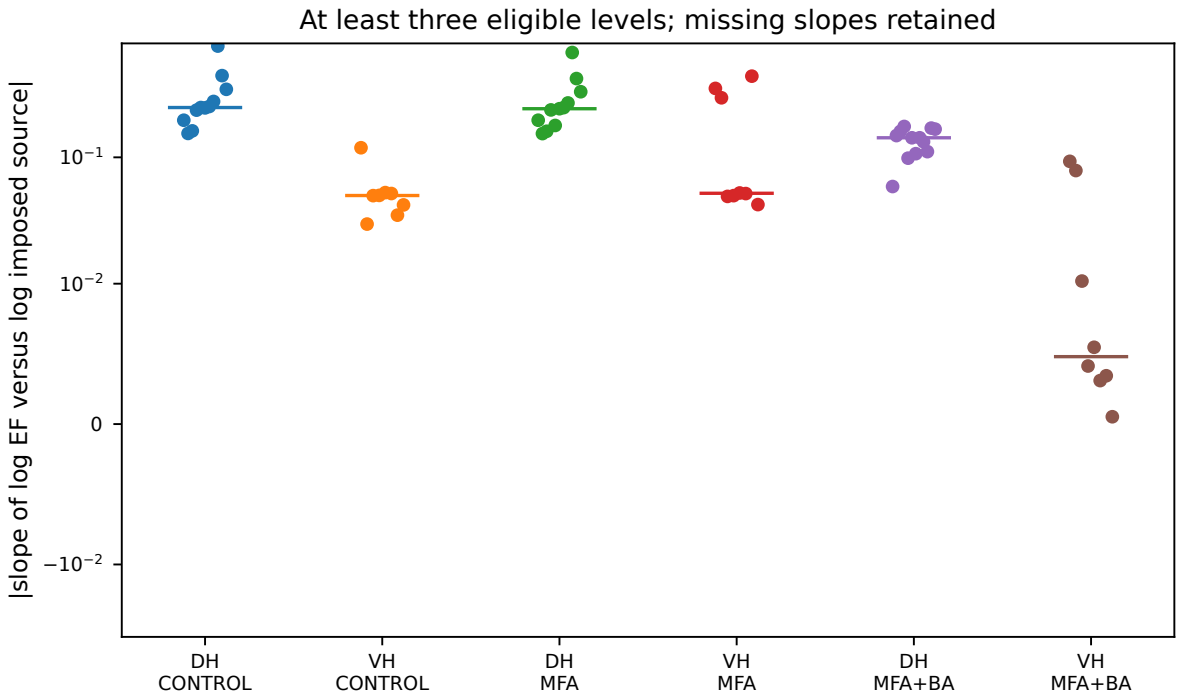


Figure D40: Absolute slope of $\log EF$ **against the log imposed input.** Earlier analysis. Signed and absolute sensitivities are kept separate, and the load is varied within each model. The slope against observed amplitude corresponds to the definition used for the tissue analysis (Appendix B); the sensitivity to the imposed input has a different predictor. A flat response can coexist with a compromised mechanism.

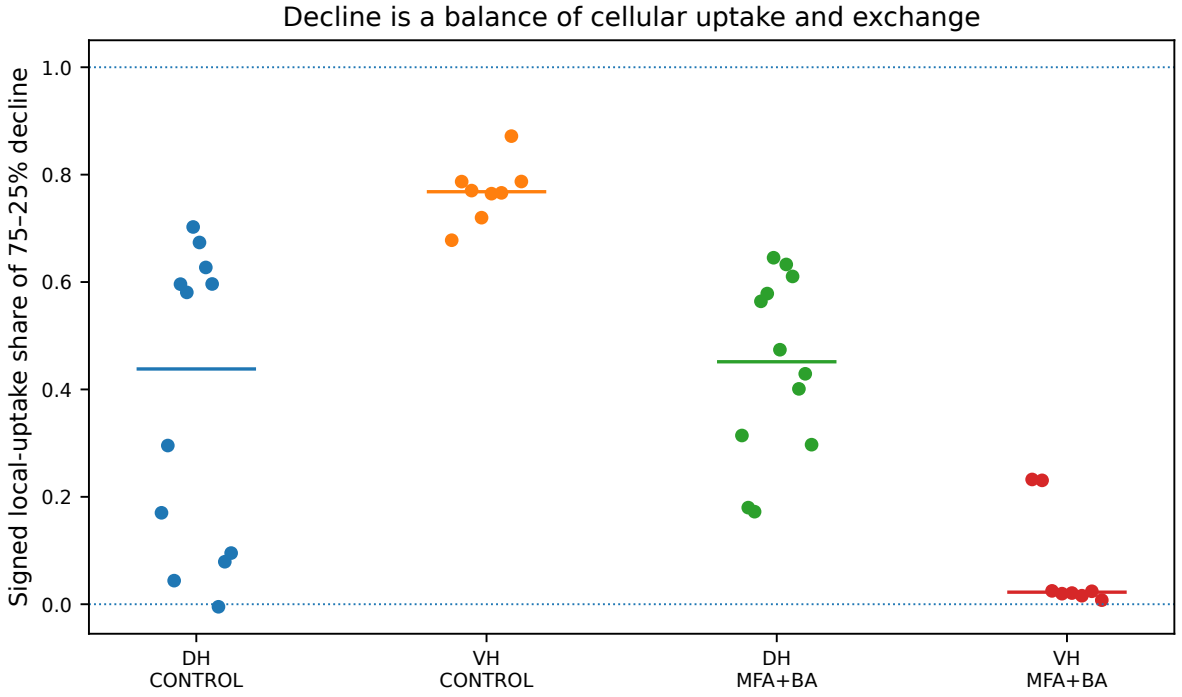


Figure D41: The mechanism of recovery changes under intervention. Earlier analysis. The fraction is the signed integral of local extracellular removal divided by the net removal over the same recovery interval; values below zero or above one indicate opposing fluxes and are not clipped. Exchange with the reservoir can keep EF stable when local uptake is strongly reduced.

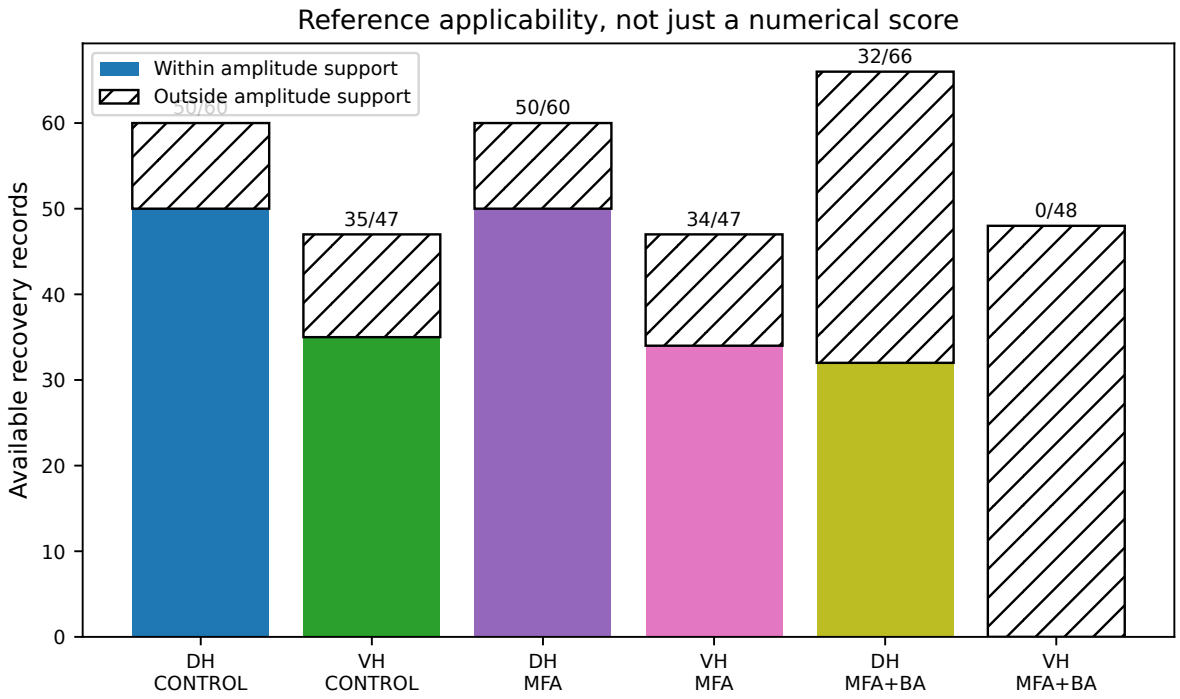


Figure D42: Comparison with the model reference requires overlap. Earlier analysis. The CONTROL-only reference excludes the tested model. Hatched entries lie outside the amplitude range of the reference for the same region and load; they are flagged extrapolations, not experimentally calibrated resilience. Missing recoveries are counted separately.

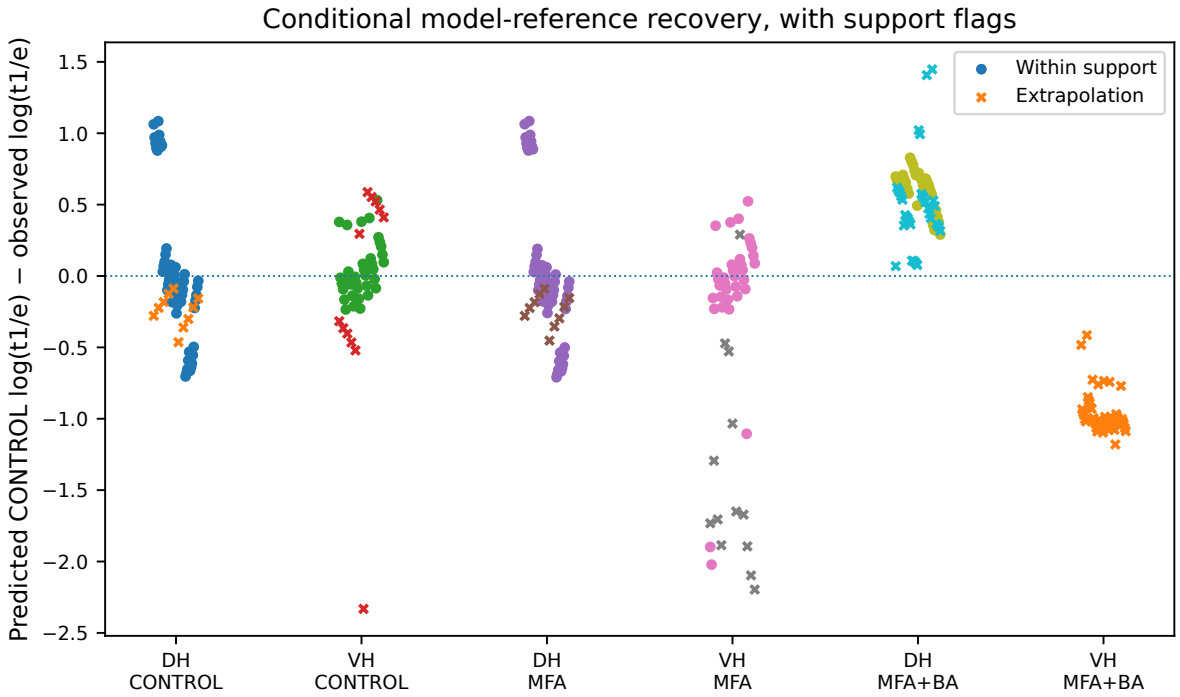


Figure D43: Recovery residuals relative to the model reference. Earlier analysis. Positive values indicate recovery faster than the CONTROL-model prediction at the given amplitude and load. The experimental healthy-tissue reference was not used. Residuals outside the range of the reference are not treated as validated resilience comparisons. Points are repeated model–load records, not independent cells.

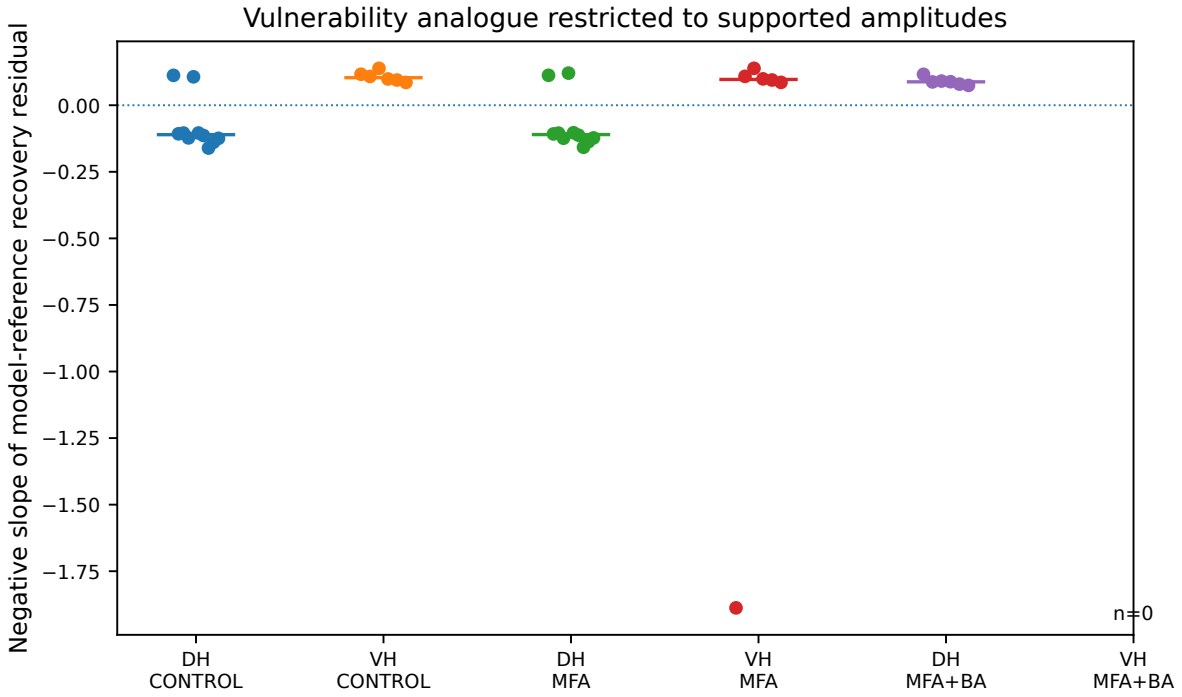


Figure D44: Vulnerability slopes relative to the model reference. Earlier analysis. At least three recovery observations within the reference range are required; an empty group is not zero vulnerability. This is a model analogue, not an animal-level or seizure-risk quantity.

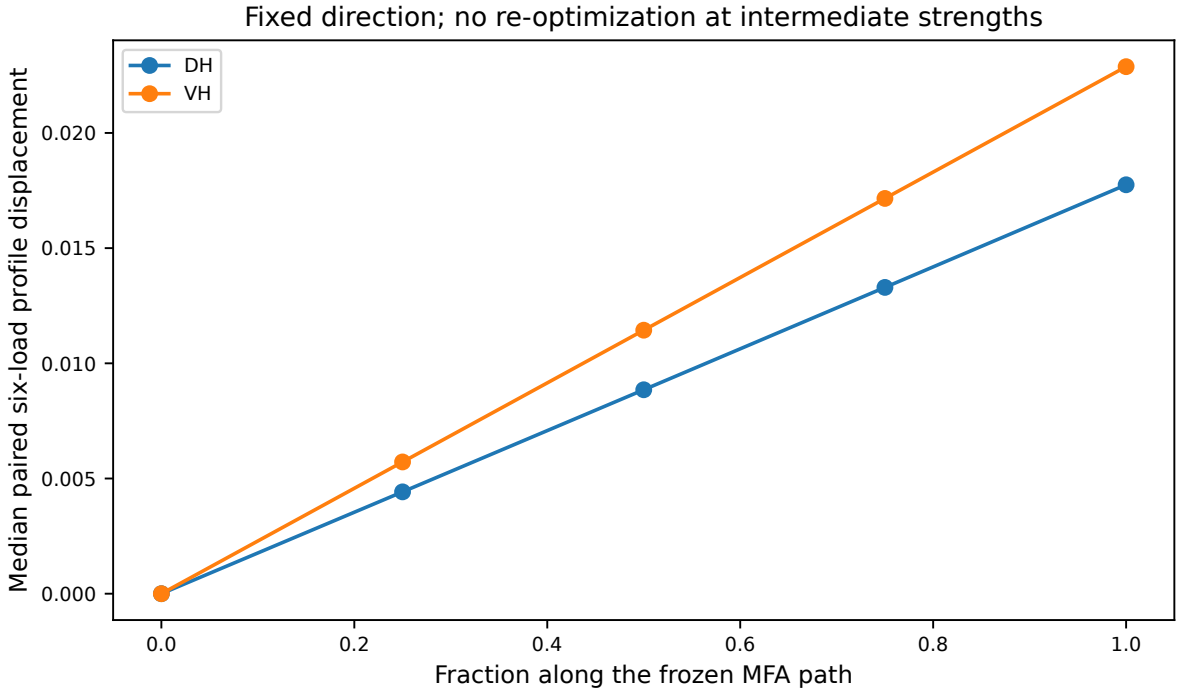


Figure D45: Graded MFA-like intervention. Earlier analysis. Multiplicative factors are interpolated logarithmically from identity and the midpoint shift linearly. The curve tests the strength of the effective intervention, not the drug concentration. "Frozen MFA path on the panel denotes the fixed MFA-like direction.

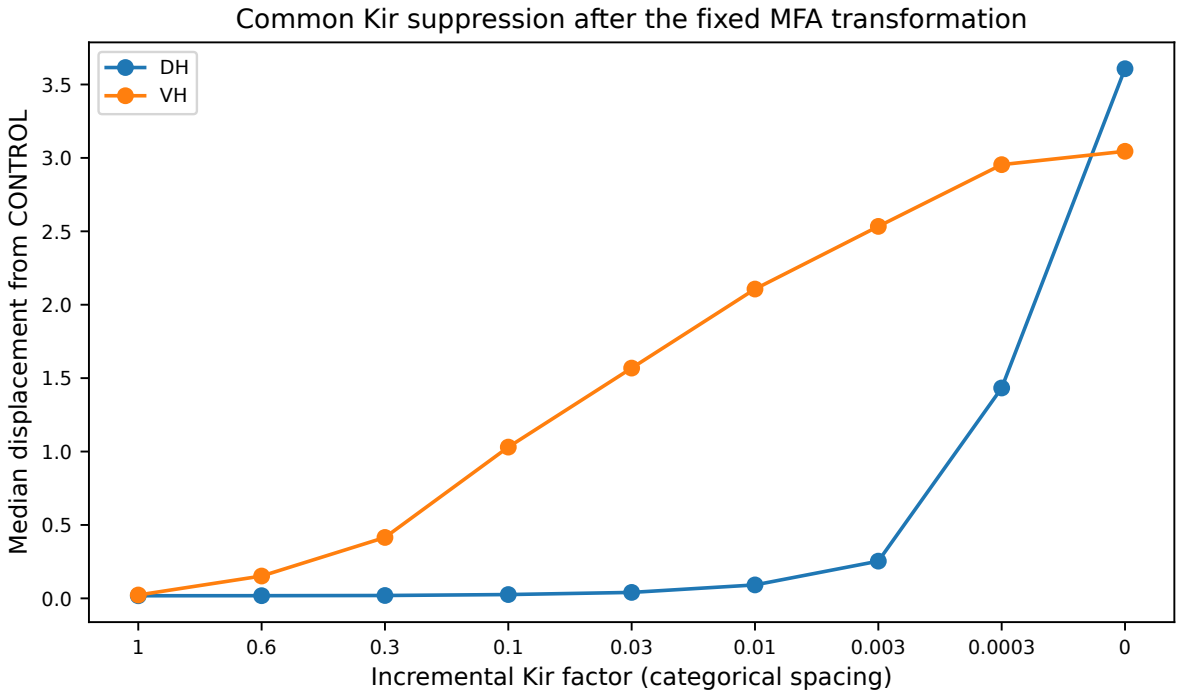


Figure D46: Graded Kir reduction. Earlier analysis. The value at factor one is the MFA-like condition under the same input and initialization. Complete removal of Kir is evaluated separately, and the Kir factor was not reselected on the potassium outcome.

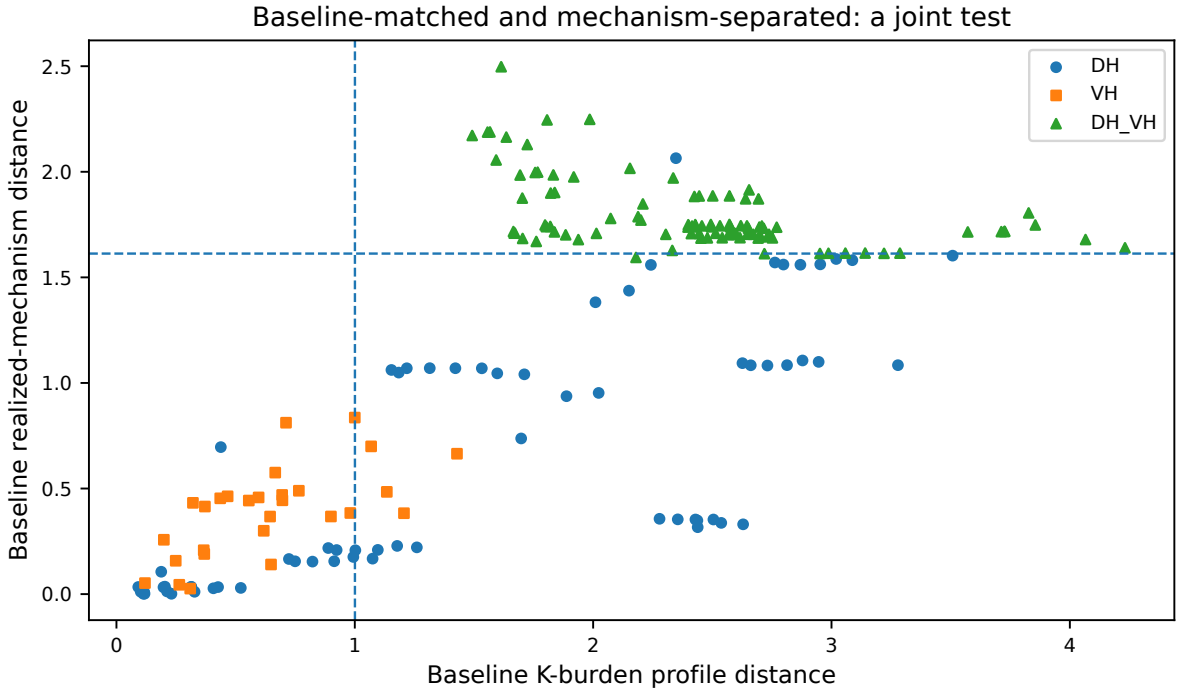


Figure D47: Baseline-matched, mechanism-separated pair test. Earlier analysis. The target region is a potassium distance at or below the lower quartile of baseline distances together with a mechanism distance at or above the median. No pair meets both criteria in this twenty-model panel with common input, and a more permissive variant is not counted as meeting the primary criterion. This does not invalidate the voltage-based degeneracy shown in the larger repertoire of the article.

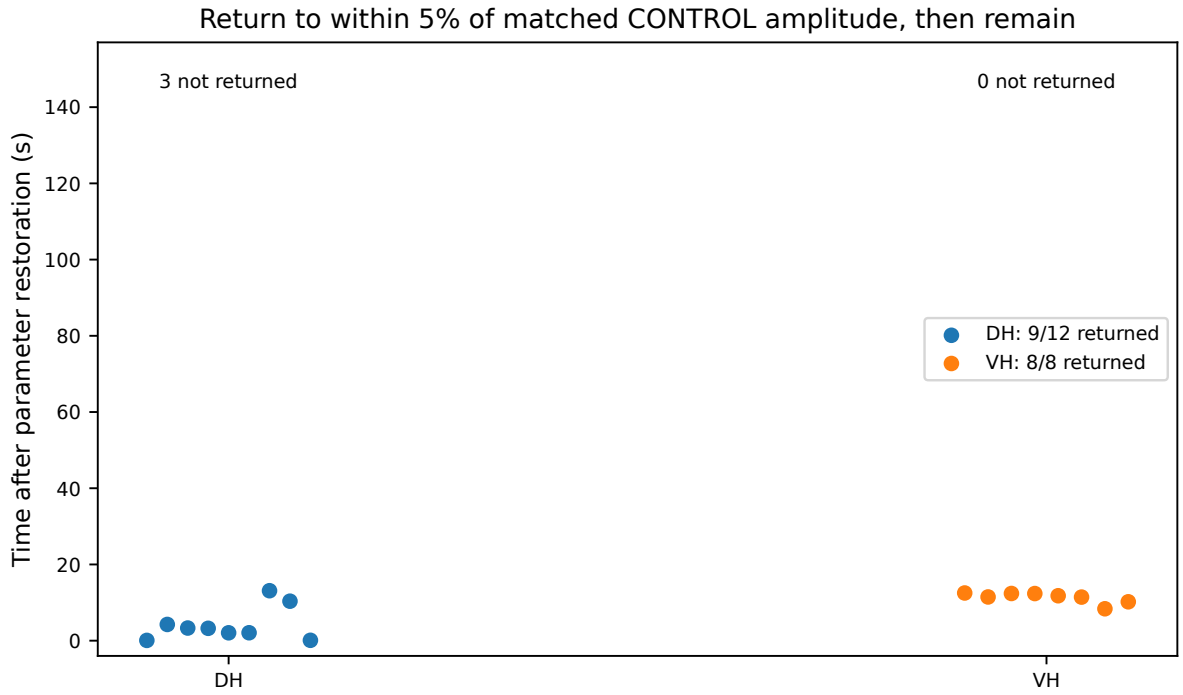


Figure D48: Recovery after parameters are restored without resetting the state. Earlier analysis. The perturbation starts from a common state initialized under CONTROL and is removed at input offset; the reference is the matching unperturbed trajectory. Models that do not return within 150 s are retained, and medians among the models that return should be read with their counts. This tests the return of the output within a finite time, not the stability of the full state.

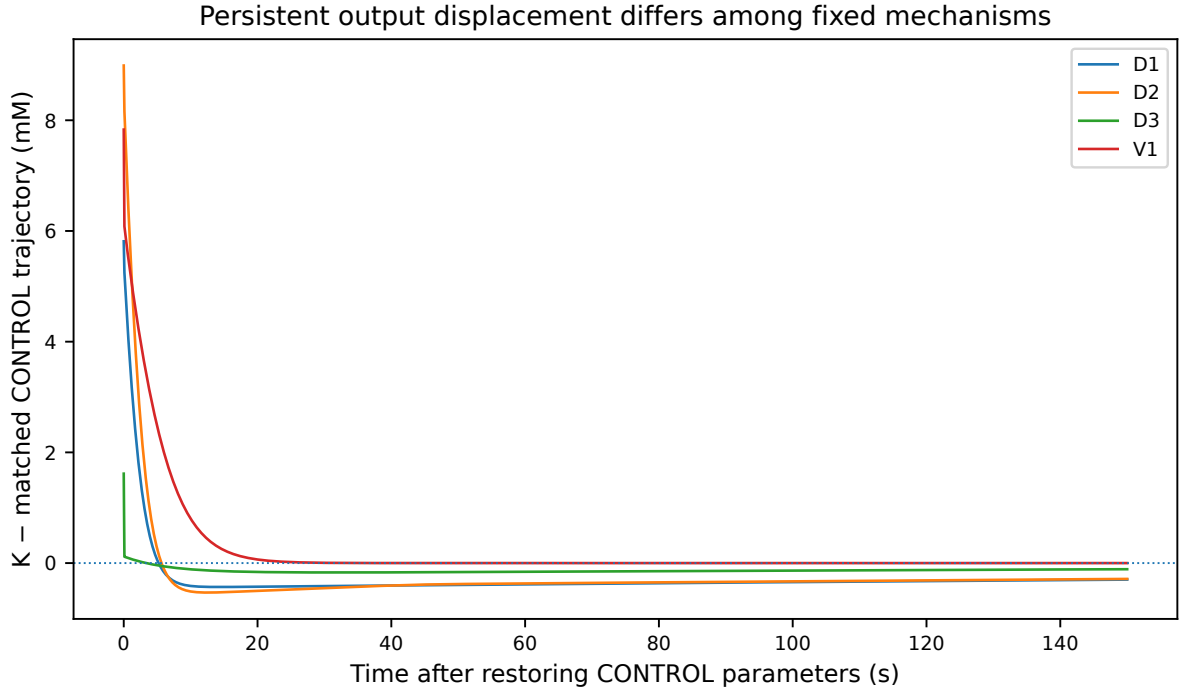


Figure D49: Restoration with the state preserved: examples. Earlier analysis. D1, D2, D3 and V1 are the four jointly supported models; they were not selected on their response here. Restoring the parameters does not erase the state accumulated during the intervention.

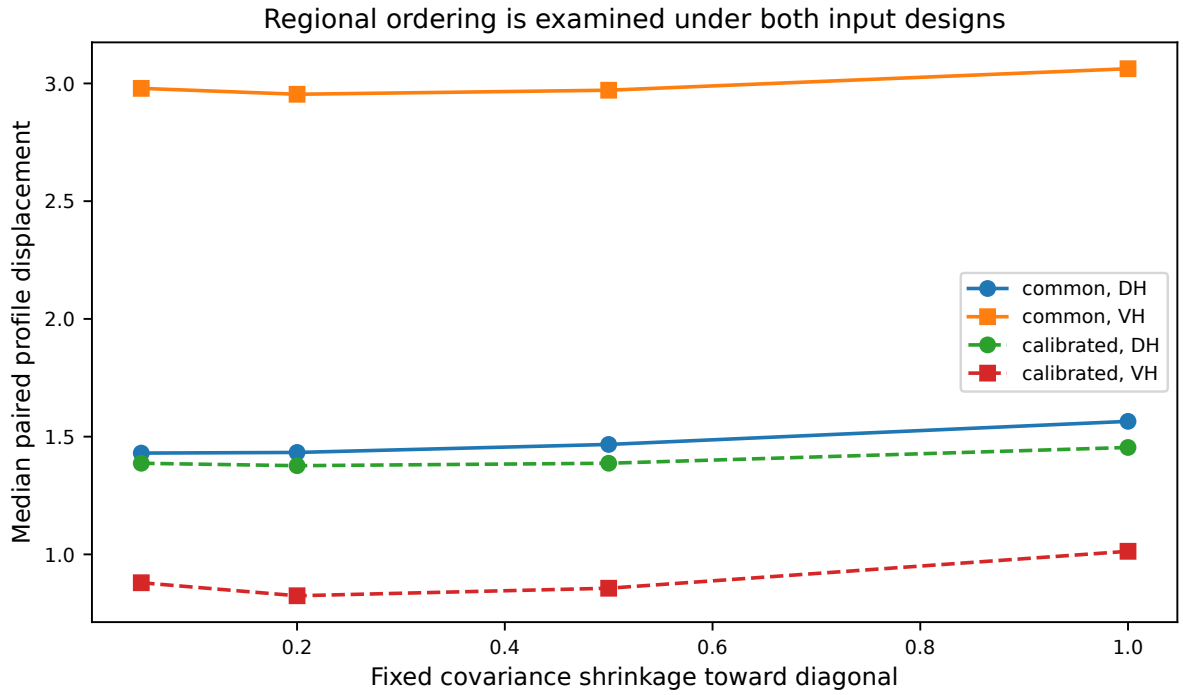


Figure D50: Sensitivity to the metric and the protocol. Earlier analysis. The common 30-s input and the original CONTROL-calibrated protocol each use their own baseline metric, so absolute distances are not comparable between protocols, and the ordering of models differs between them. Shrinkage was not tuned to favour an ordering.

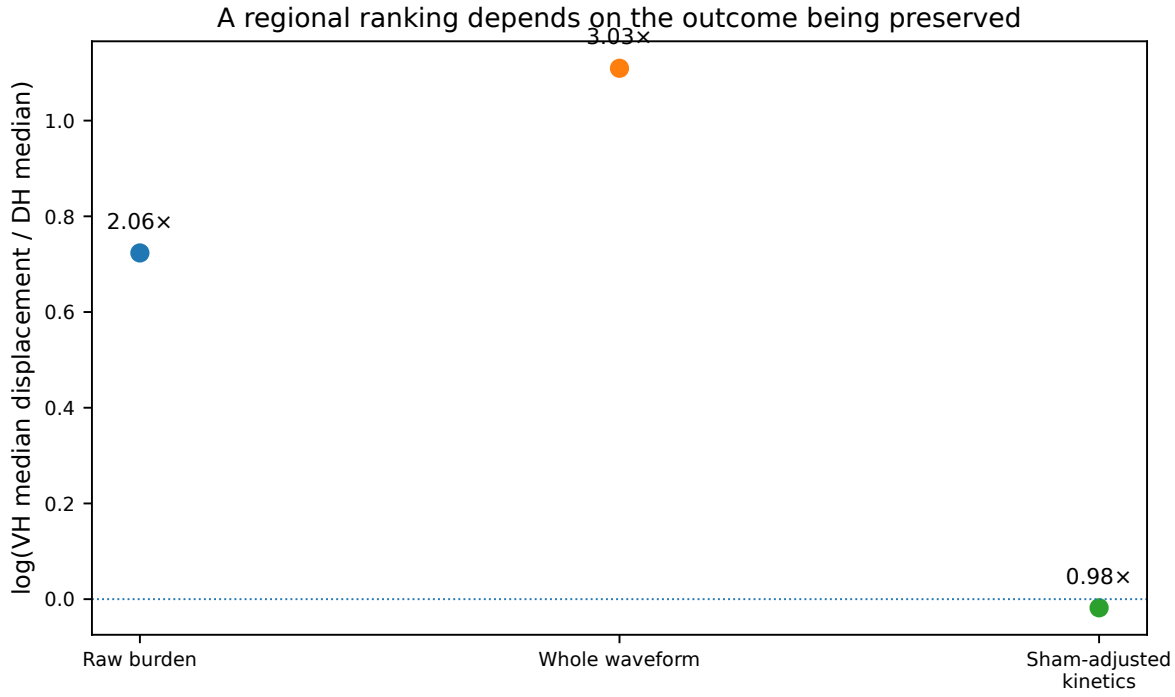


Figure D51: Preservation of raw and input-evoked profiles. Earlier analysis. Each ratio compares the regions within one metric. The no-input-adjusted five-feature profile measures the deviation from the matched no-input trajectory, not the absolute extracellular concentration; its six-load profiles are available for all twenty models and do not replace the raw cases without recovery.

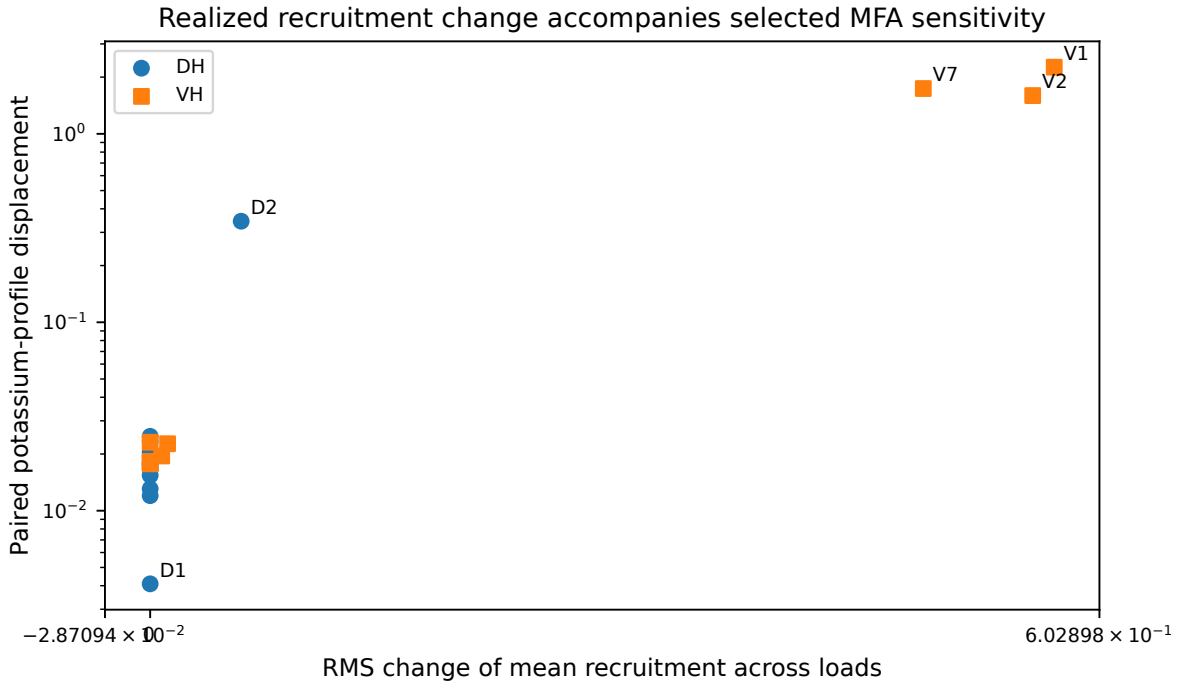


Figure D52: Changes in output and recruitment. Earlier analysis. The association is descriptive across fixed parameter sets and does not isolate the causal effect of any single parameter. It relates the heterogeneity of the responses to the realized changes in the model mechanisms.

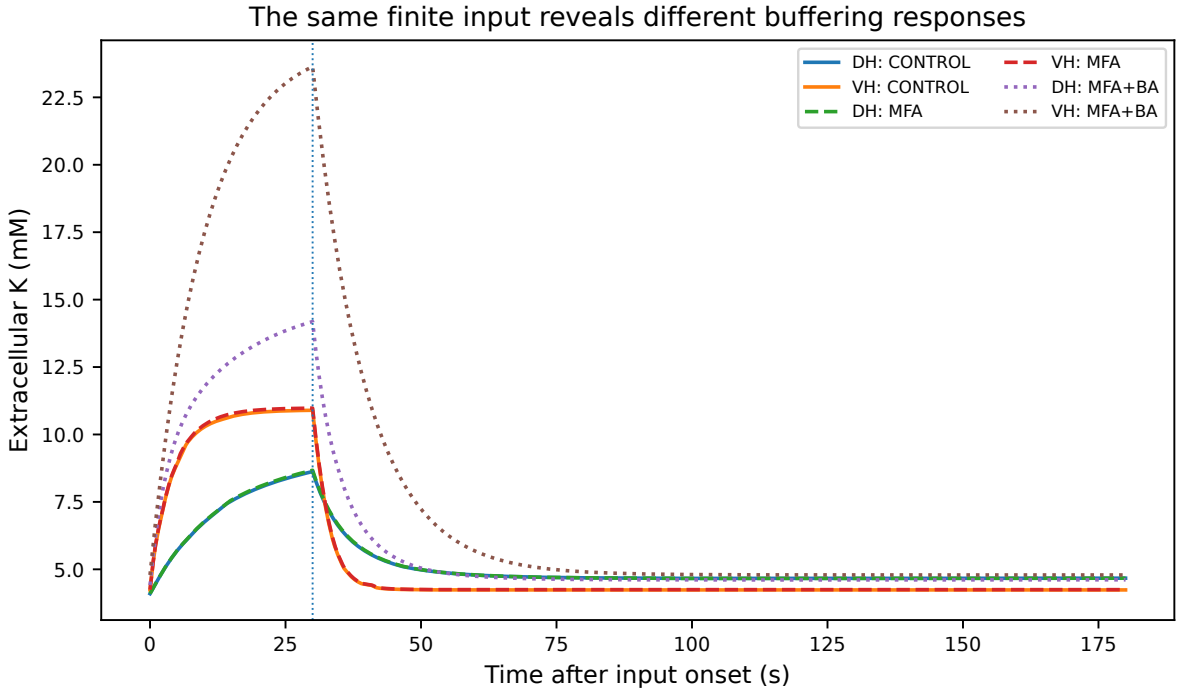


Figure D53: Potassium waveforms under the same input. Earlier analysis. Regional median trajectories at the 25-mM reservoir; the plotted traces end at 30 s. Initializations are specific to each condition in the main experiment; the paired no-input and common-initialization restoration analyses examine that choice separately.

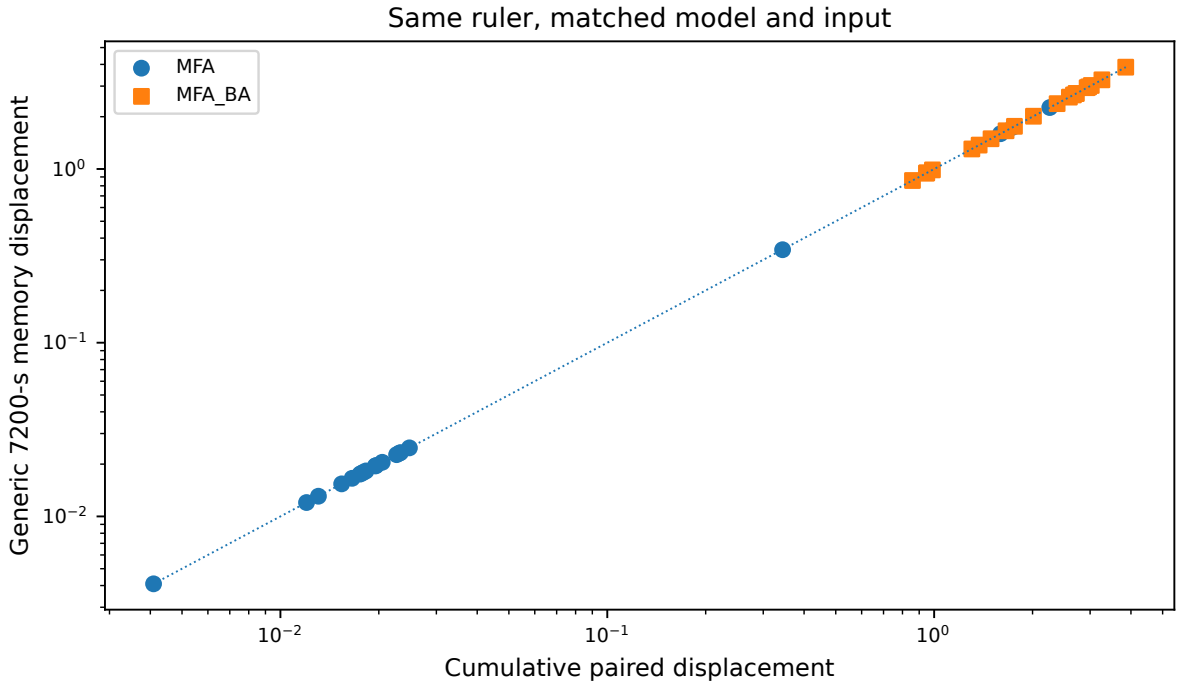


Figure D54: Sensitivity to the model formulation. Earlier analysis. Each point is a model–condition pair with the same primitive parameters, the same input and the CONTROL metric of the cumulative formulation. The similarity applies to these protocols and is not assumed to hold generally.

Transformations to other times of day

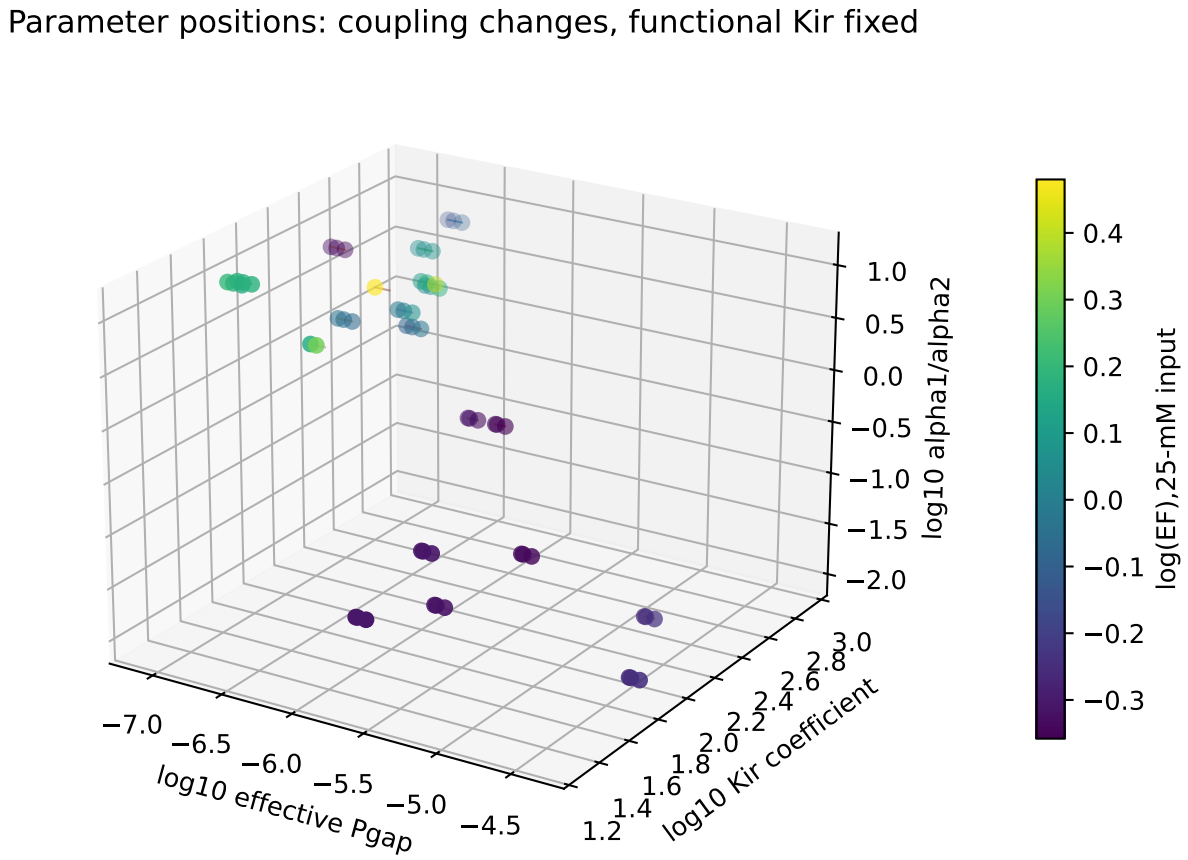


Figure D55: Three-dimensional parameter projection of the time-of-day transformations. The primary transformation maps the ratios of dye-coupled cell counts onto $P_{K,\mathrm{gap}}$ only, with Kir conductance, conversion ratio and gate fixed. Colour shows the simulated EF, not a measured molecular property.

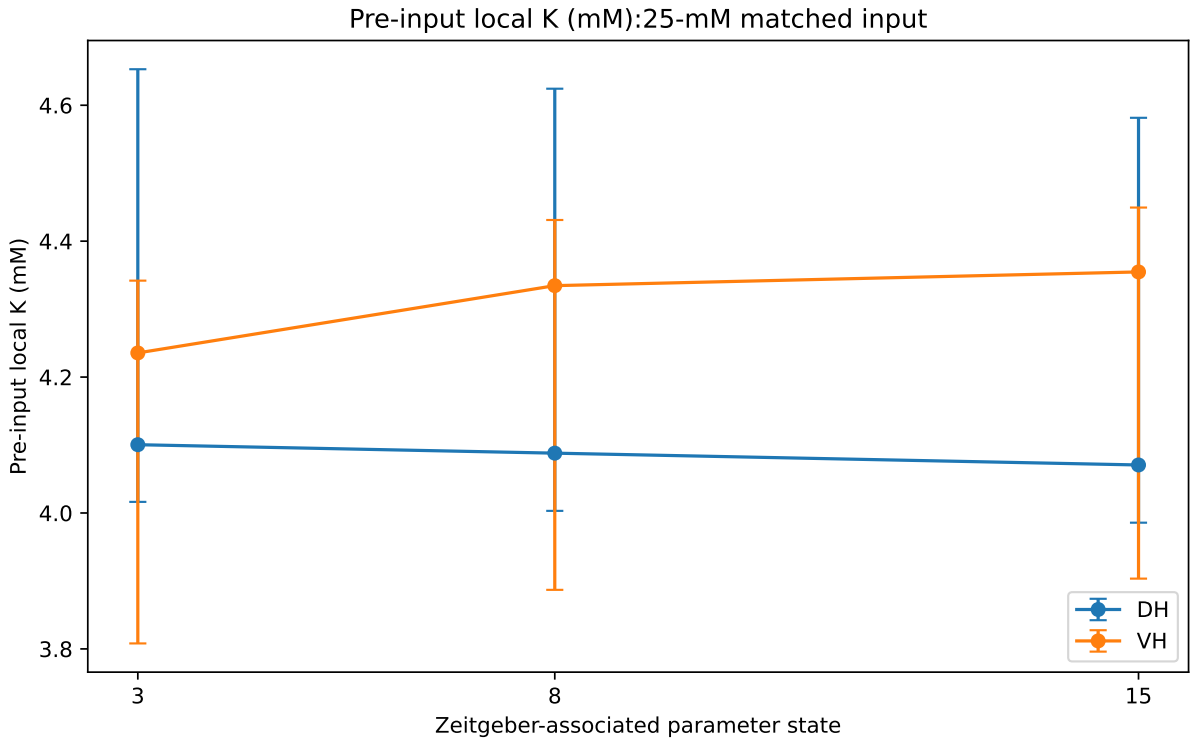


Figure D56: Pre-input extracellular potassium across times of day. Median and interquartile range across models, not animal confidence intervals. The values follow from the assumed mapping of dye coupling onto $P_{K,\mathrm{gap}}$. The nominal reservoir concentration remains 4.8 mM; the pre-input concentration reflects the finite model preparation.

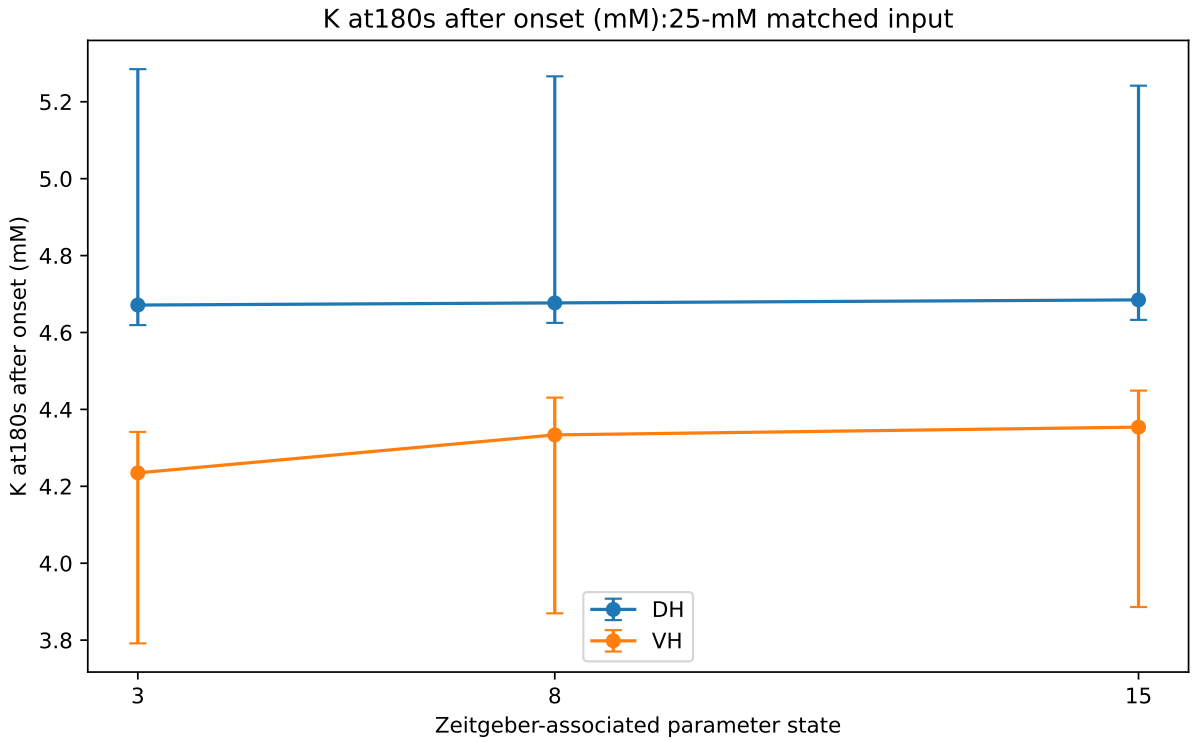


Figure D57: Potassium 180 s after input onset across times of day. Median and interquartile range across models, not animal confidence intervals. The values follow from the assumed mapping of dye coupling onto $P_{K,\mathrm{gap}}$. The nominal reservoir concentration remains 4.8 mM; the pre-input concentration reflects the finite model preparation.

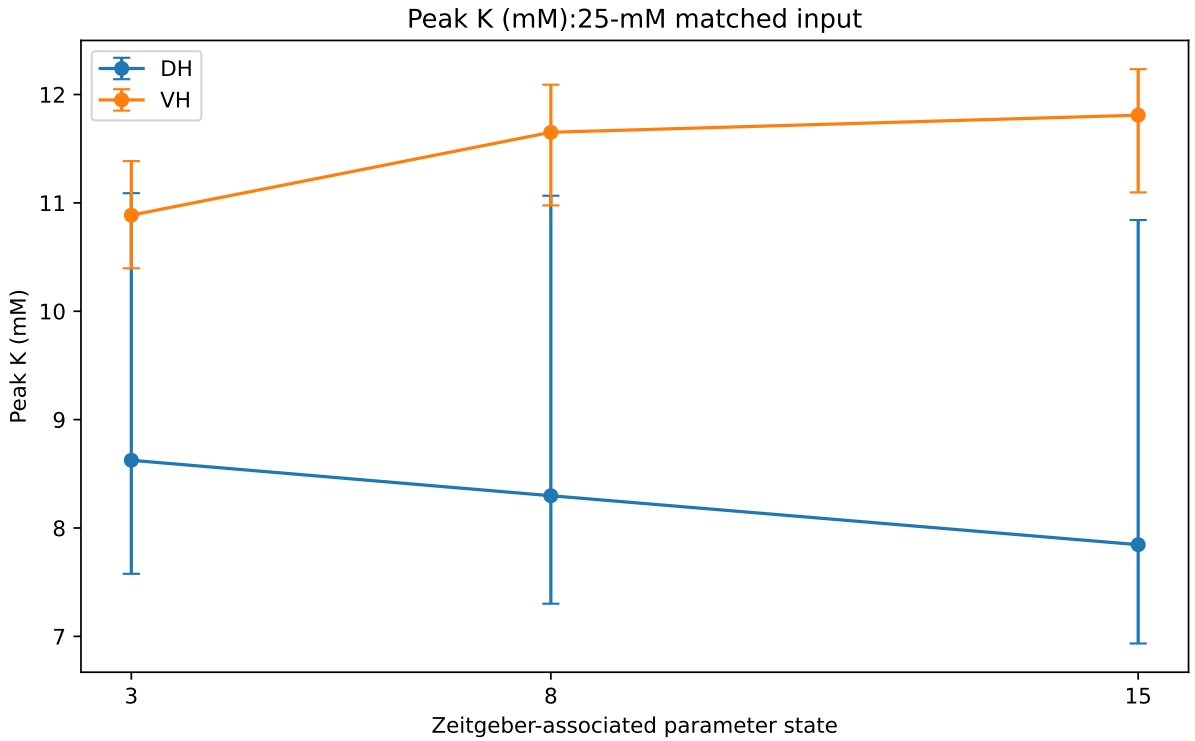


Figure D58: Peak potassium across times of day. Median and interquartile range across models, not animal confidence intervals. The values follow from the assumed mapping of dye coupling onto $P_{K,\mathrm{gap}}$. The nominal reservoir concentration remains 4.8 mM; the pre-input concentration reflects the finite model preparation.

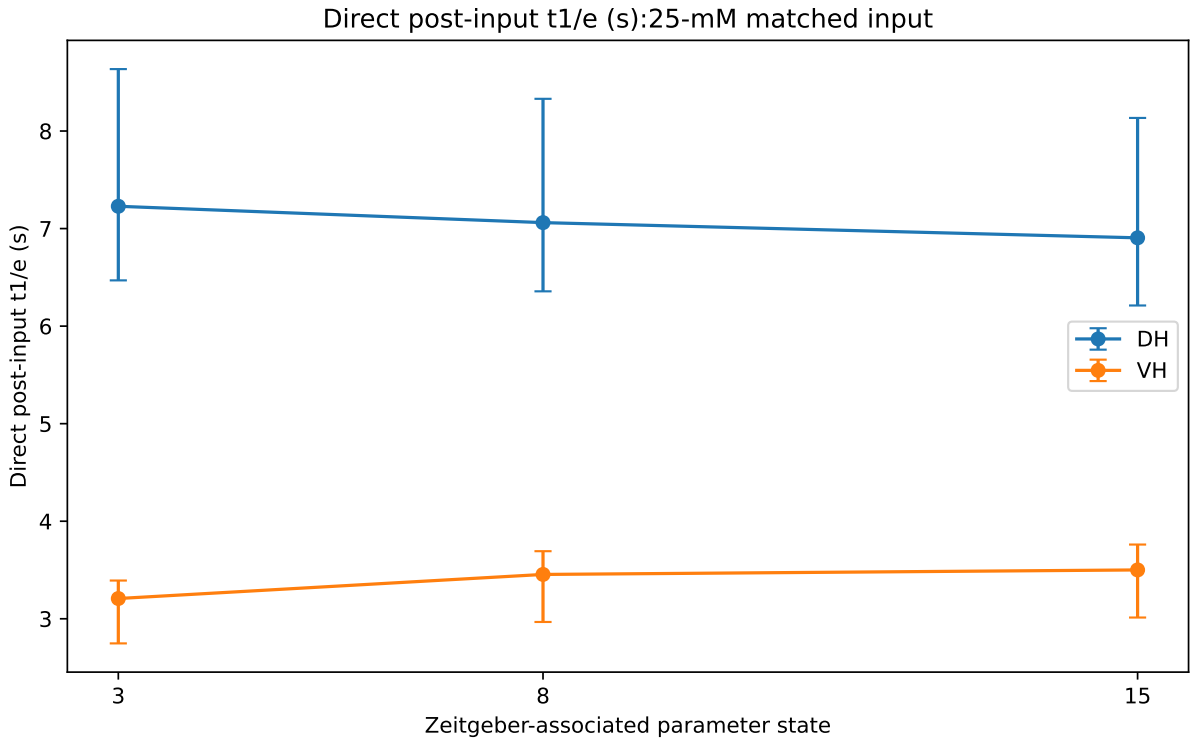


Figure D59: Direct post-input $t_{1/e}$ across times of day. Median and interquartile range across models, not animal confidence intervals. The values follow from the assumed mapping of dye coupling onto $P_{K,\mathrm{gap}}$. The nominal reservoir concentration remains 4.8 mM; the pre-input concentration reflects the finite model preparation.

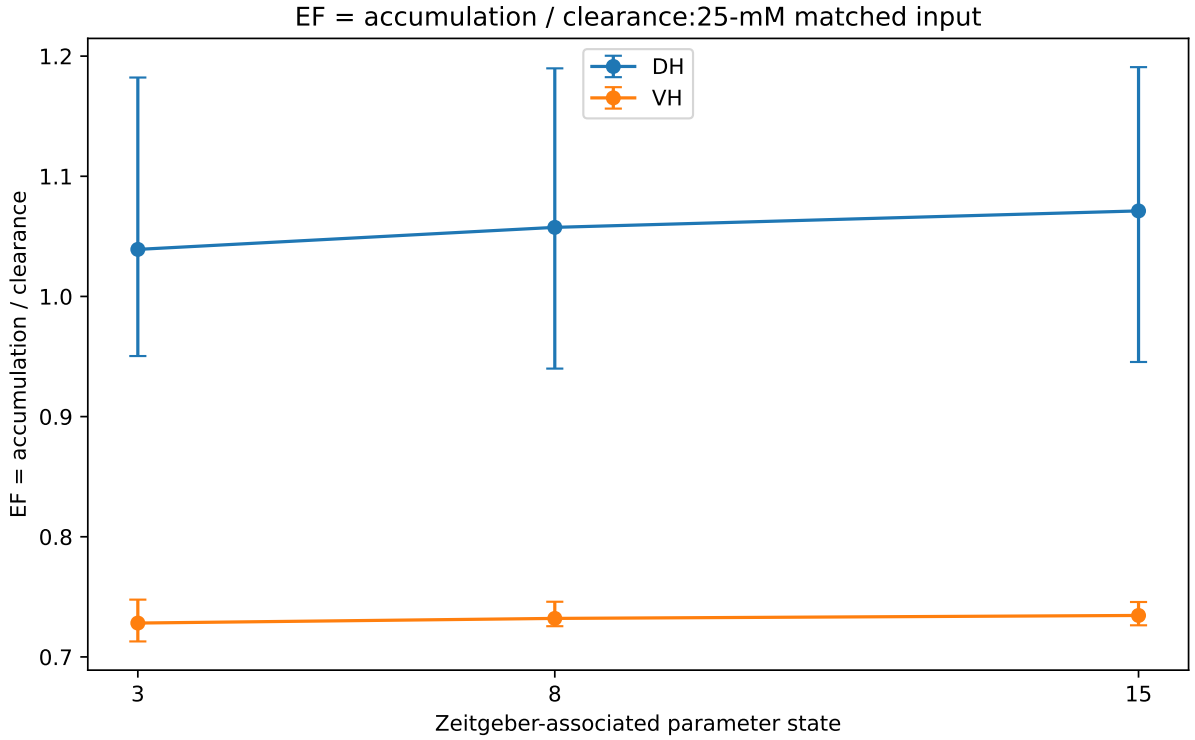


Figure D60: Accumulation–recovery balance (EF) across times of day. Median and interquartile range across models, not animal confidence intervals. The values follow from the assumed mapping of dye coupling onto $P_{K,\mathrm{gap}}$. The nominal reservoir concentration remains 4.8 mM; the pre-input concentration reflects the finite model preparation.

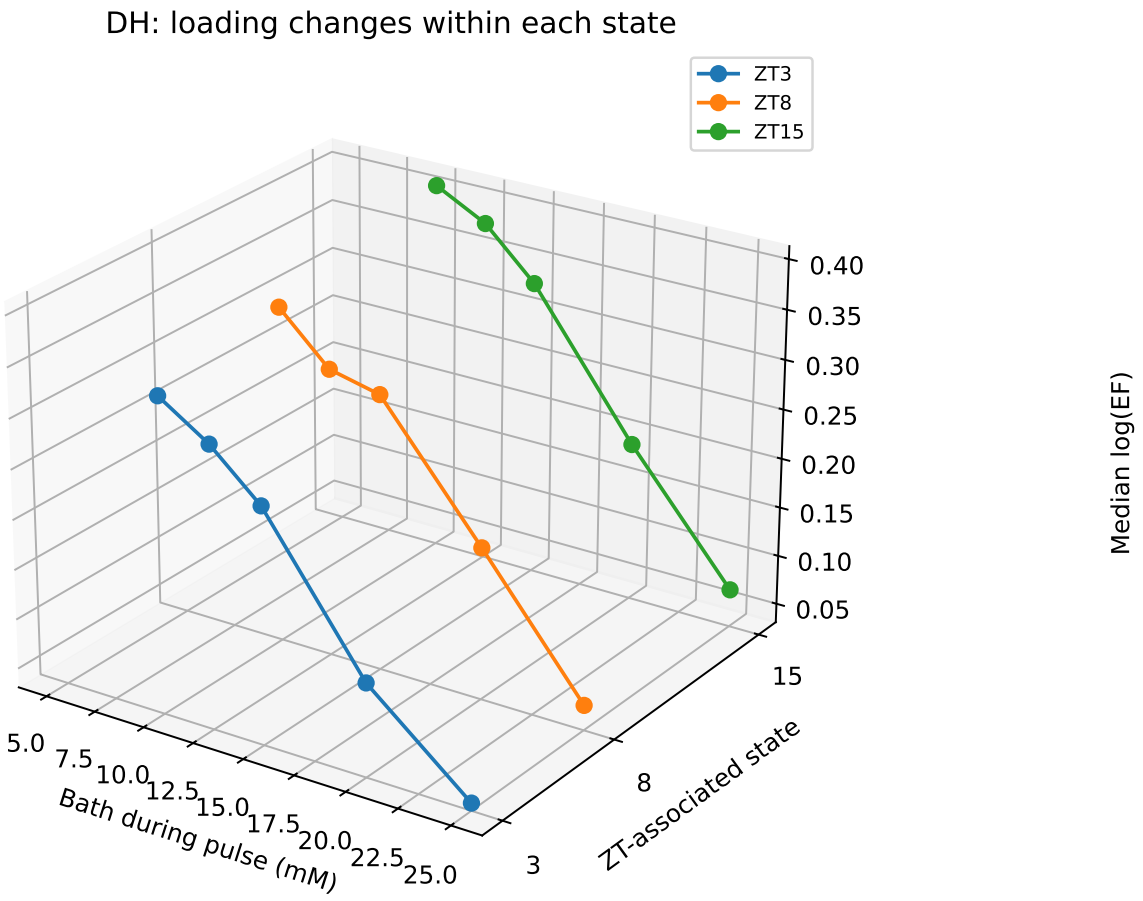


Figure D61: EF across load in the DH-like models at three times of day. Lines connect the six tested loads within each of the three time-of-day states; no continuous surface is interpolated. Missing values are reported per model.

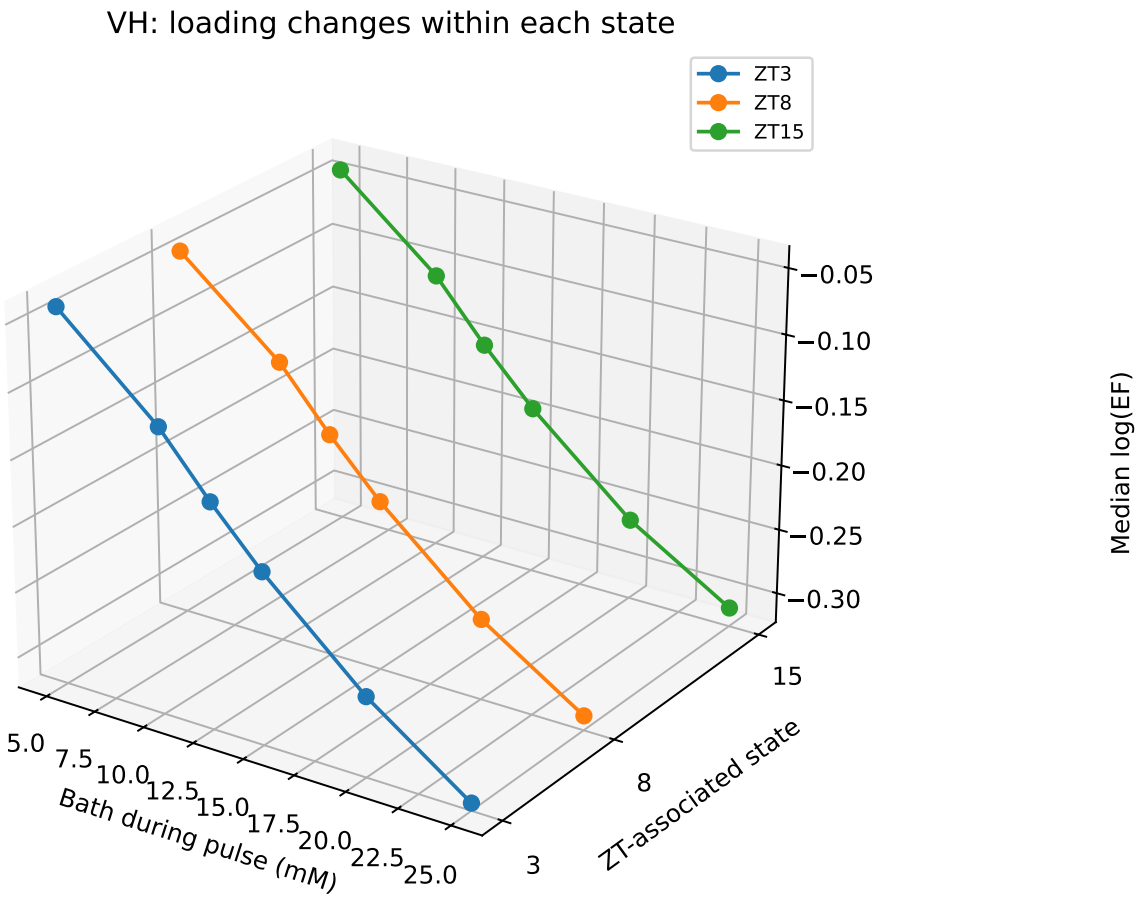


Figure D62: EF across load in the VH-like models at three times of day. Lines connect the six tested loads within each of the three time-of-day states; no continuous surface is interpolated. Missing values are reported per model.

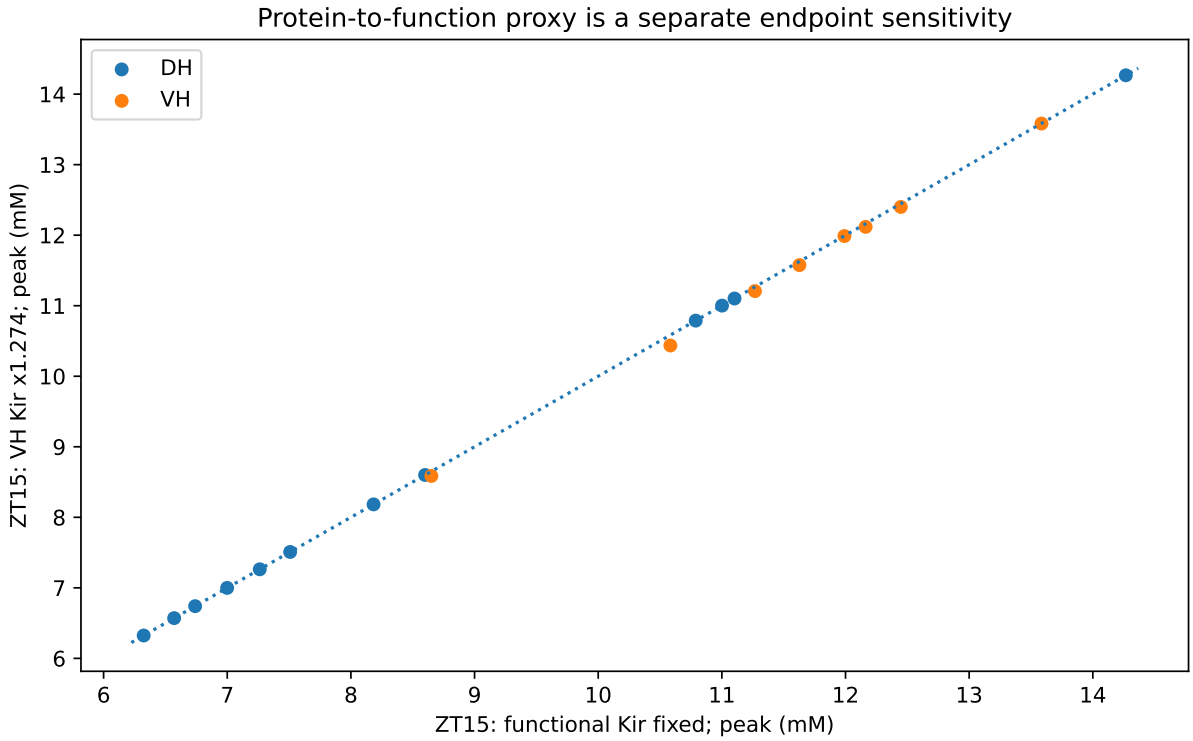


Figure D63: Sensitivity to a protein-based change in Kir. At ZT15 only, the reported 27.4% increase in normalized Kir4.1 labelling in VH relative to ZT3 (Kiani et al., 2026) is applied hypothetically to g_{Kir}. The primary transformation keeps the Kir conductance fixed because no time-of-day effect on the current was detected. No ZT8 protein factor is assumed.

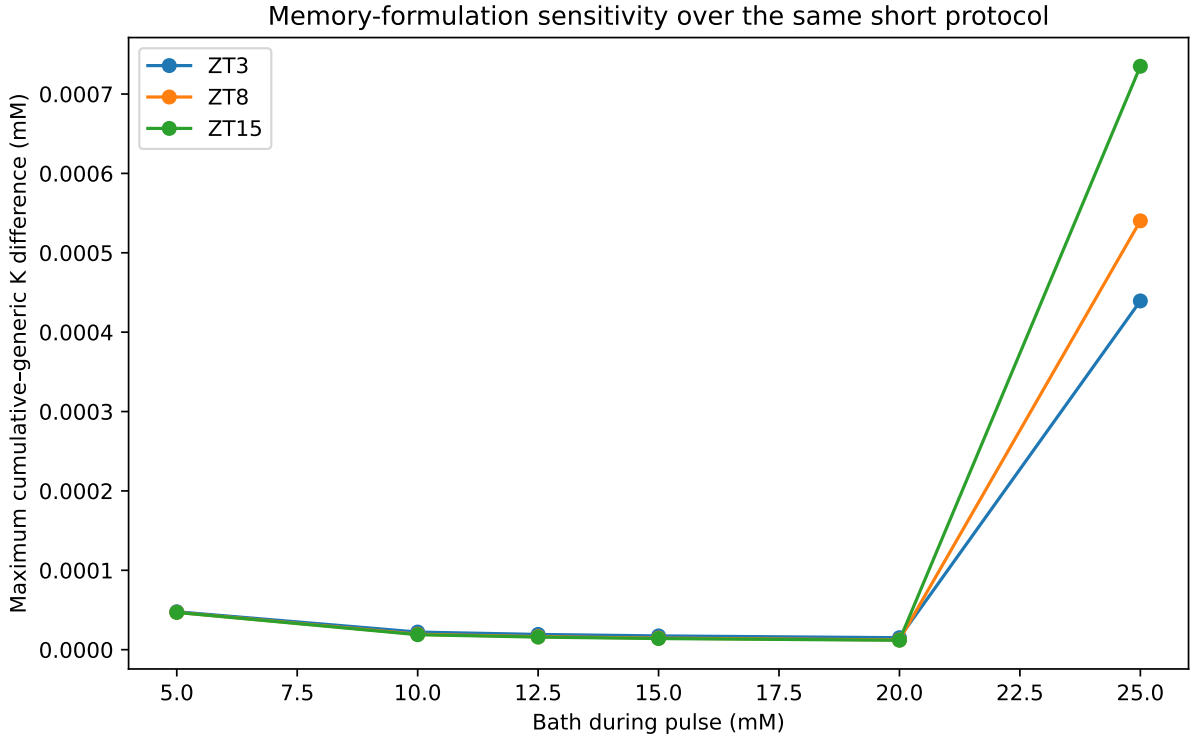


Figure D64: Cumulative versus generic formulation across times of day. Maximum difference over the full simulated trace, across models at each load; generic memory 7,200 s. The comparison applies to this protocol and is not a general equivalence.

Propagated measurement uncertainty

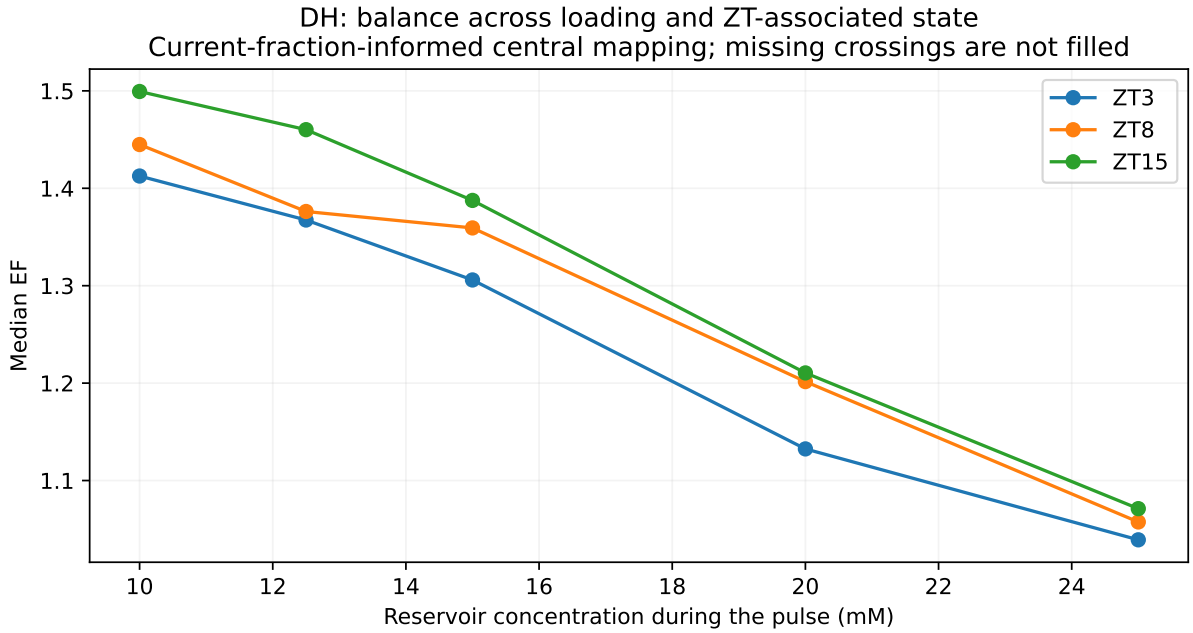


Figure D65: EF across load and time of day in the DH-like models. Six-level model challenge with propagated tracer uncertainty; it does not correspond to the five stimulation protocols of the tissue dataset.

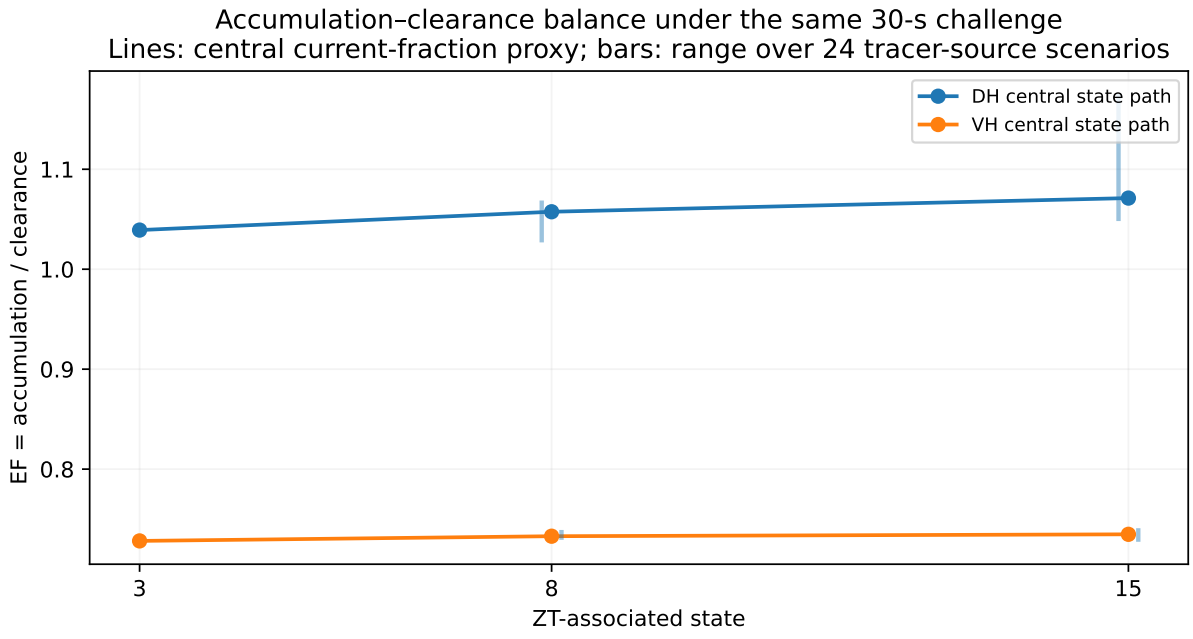


Figure D66: Accumulation–recovery balance with propagated tracer uncertainty. Ranges are the ranges of the regional model medians across the sampled coupling ratios, not confidence intervals or biological cell variability. The tracer-source draws named on the panel are these sampled ratios; the lines use the central current-fraction factor.

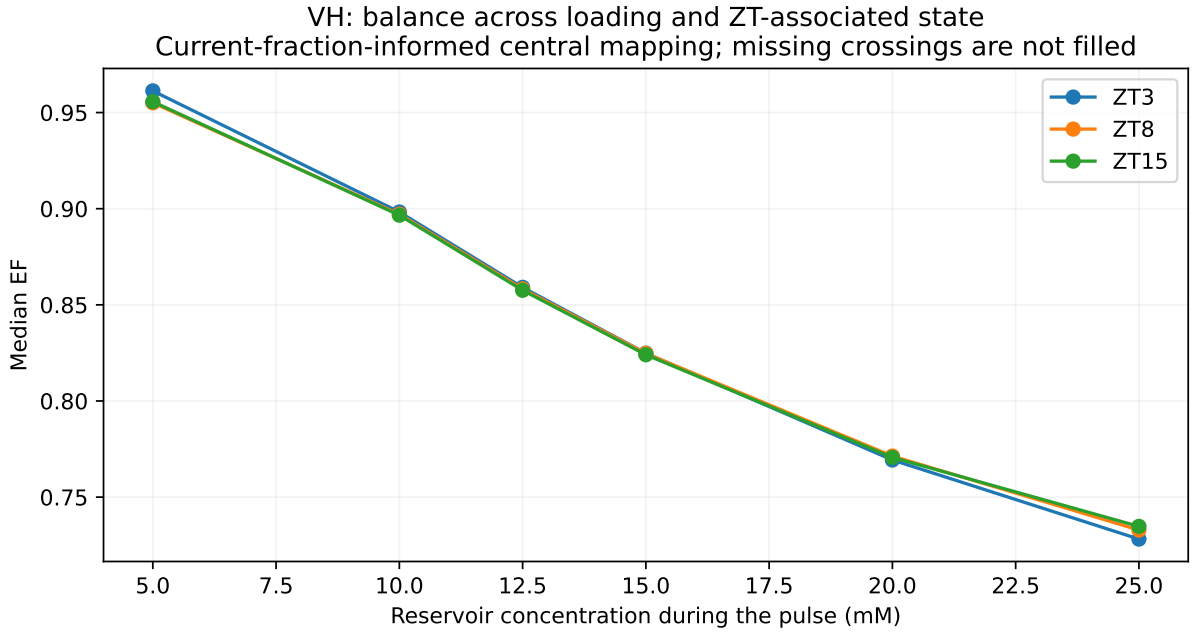


Figure D67: EF across load and time of day in the VH-like models. Six-level model challenge with propagated tracer uncertainty; it does not correspond to the five stimulation protocols of the tissue dataset.

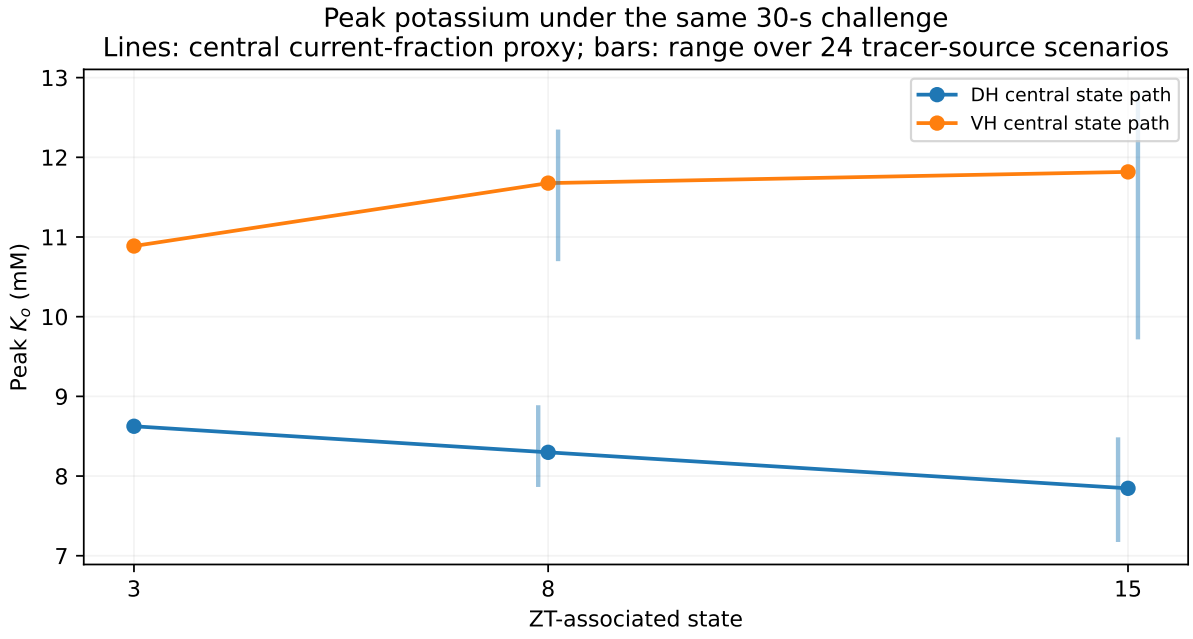


Figure D68: Peak potassium with propagated tracer uncertainty. Ranges are the ranges of the regional model medians across the sampled coupling ratios, not confidence intervals or biological cell variability. The tracer-source draws named on the panel are these sampled ratios; the lines use the central current-fraction factor.

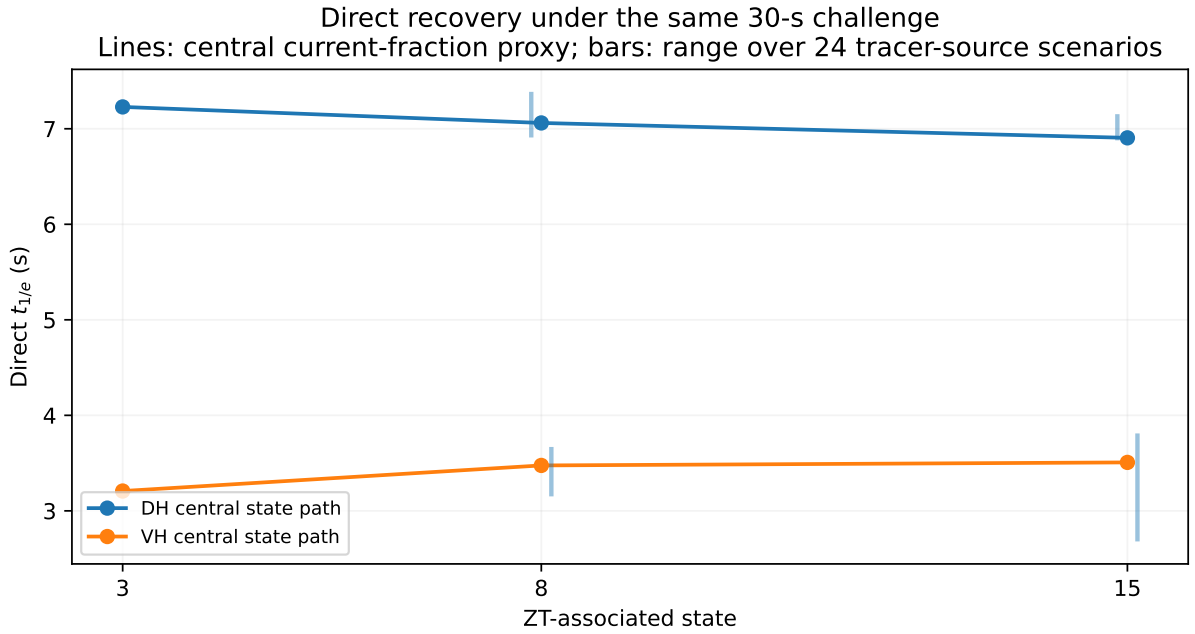


Figure D69: Direct recovery time with propagated tracer uncertainty. Ranges are the ranges of the regional model medians across the sampled coupling ratios, not confidence intervals or biological cell variability. The tracer-source draws named on the panel are these sampled ratios; the lines use the central current-fraction factor.